Online Resource 2 for

An exact line-integral formulation for evaluating tesseroid potentials and their derivatives through third order

Extended mathematical derivation and numerical implementation

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Introduction

This online resource provides the complete mathematical derivation and implementation details supporting the gauge-optimized tesseroid line-integral method. It establishes the dimensional reduction, regularity and distributional scope, cancellation-resistant kernels, observation-coordinate derivatives through third order, projection-aware adaptive quadrature, and the three-region hybrid used for gridded shells. Validation figures and statistical audits are supplied separately in Online Resource 1 (`ESM_1.pdf`).

Exact reduction of the gravitational potential of a homogeneous tesseroid to two paired one-dimensional integrals, with derivatives through third order

The dimensional reduction is obtained with a gauge-optimized divergence field: a divergence-free curl converts the angularly varying flux on every source sphere into its complete-sphere mean and transfers the residual exactly to the four side faces. A single three-dimensional divergence theorem then yields the explicit term and four line integrals, which pair into two.

1. Geometry and notation

Let the observation point and source point be

$$P = P(r, \varphi, \lambda), \qquad Q = Q(r', \varphi', \lambda'), \tag{1}$$

where φ and φ' are geocentric latitudes. Consider the homogeneous tesseroid

$$\Omega = [r_1, r_2] \times [\varphi_1, \varphi_2] \times [\lambda_1, \lambda_2], \tag{2}$$

with

$$0 < r_1 < r_2, \qquad -\frac{\pi}{2} \le \varphi_1 < \varphi_2 \le \frac{\pi}{2}, \qquad -\pi \le \lambda_1 < \lambda_2 \le \pi. \tag{3}$$

The longitude bounds are principal longitudes. A tesseroid that crosses the antimeridian is not represented by a wrapped interval. It must first be partitioned into two non-wrapping tesseroids with longitude intervals $[\lambda_1, \pi]$ and $[-\pi, \lambda_2]$; their contributions are then evaluated and summed separately. The complete interval $[-\pi, \pi]$ remains admissible. The density ρ is constant.

Define

$$\beta = \lambda' - \lambda, \tag{4}$$

$$c = \cos\psi = \sin\varphi \sin\varphi' + \cos\varphi \cos\varphi' \cos\beta, \tag{5}$$

and

$$l = l(r'; c) = \sqrt{r^2 + r'^2 - 2rr'c}. \tag{6}$$

The Newtonian gravitational potential is

$$V(r, \varphi, \lambda) = G\rho \int_{\lambda_1}^{\lambda_2} \int_{\varphi_1}^{\varphi_2} \int_{r_1}^{r_2} \frac{r'^2 \cos\varphi'}{l} \, \mathrm{d}r' \mathrm{d}\varphi' \mathrm{d}\lambda'. \tag{7}$$

Throughout the derivation, the observation point is assumed not to lie on $\partial\Omega$. It may lie either inside or outside Ω.

At the source point, let $\boldsymbol{e}_{r'}$, $\boldsymbol{e}_{\varphi'}$, and $\boldsymbol{e}_{\lambda'}$ denote the usual orthonormal spherical basis. Introduce

$$C_\varphi := \frac{\partial c}{\partial \varphi'} = \sin\varphi \cos\varphi' - \cos\varphi \sin\varphi' \cos\beta, \tag{8}$$

$$C_\lambda := \frac{1}{\cos\varphi'} \frac{\partial c}{\partial \lambda'} = -\cos\varphi \sin\beta. \tag{9}$$

If $\boldsymbol{x} = \overrightarrow{OP}$ and $\boldsymbol{y} = \overrightarrow{OQ}$, then

$$\boldsymbol{x} \cdot \boldsymbol{e}_{r'} = rc, \qquad \boldsymbol{x} \cdot \boldsymbol{e}_{\varphi'} = rC_\varphi, \qquad \boldsymbol{x} \cdot \boldsymbol{e}_{\lambda'} = rC_\lambda. \tag{10}$$

For paired lower and upper angular boundaries, define

$$\epsilon_1 = -1, \qquad \epsilon_2 = +1. \tag{11}$$

2. Exact conversion of the volume integral to the six boundary faces

This section explains the first dimensional reduction in detail. The tool is the three-dimensional divergence theorem, also called Gauss's theorem.

2.1 Statement of the divergence theorem

Let $\Omega \subset \mathbb{R}^3$ be a bounded volume whose boundary $\partial\Omega$ is piecewise smooth, and let $\boldsymbol{F}$ be continuously differentiable in a neighbourhood of $\overline{\Omega}$. If $\boldsymbol{n}$ denotes the outward unit normal on $\partial\Omega$, the divergence theorem states that

$$\boxed{\int_\Omega \nabla_{\boldsymbol{y}} \cdot \boldsymbol{F}(\boldsymbol{y})\,\mathrm{d}\Omega = \int_{\partial\Omega} \boldsymbol{F}(\boldsymbol{y}) \cdot \boldsymbol{n}\,\mathrm{d}S.} \tag{12}$$

The left-hand side integrates the local volume expansion rate $\nabla \cdot \boldsymbol{F}$ throughout the body. The right-hand side measures the net outward flux of the same vector field through the closed boundary. Equation (12) is therefore the precise mathematical statement that a volume divergence can be replaced by a boundary flux.

2.2 A normalized vector field whose divergence is the Newton kernel

Let

$$\boldsymbol{d} = \boldsymbol{y} - \boldsymbol{x}, \qquad l = \|\boldsymbol{d}\|. \tag{13}$$

We seek a vector field whose divergence is exactly $1/l$. Consider the normalized field

$$\boldsymbol{F}_0(\boldsymbol{y}) = \frac{\boldsymbol{d}}{2l} = \frac{\boldsymbol{y} - \boldsymbol{x}}{2\|\boldsymbol{y} - \boldsymbol{x}\|}. \tag{14}$$

For $\boldsymbol{y} \neq \boldsymbol{x}$, use the product rule

$$\nabla \cdot (f\boldsymbol{a}) = f\,\nabla \cdot \boldsymbol{a} + \boldsymbol{a} \cdot \nabla f \tag{15}$$

with $f = 1/l$ and $\boldsymbol{a} = \boldsymbol{d}$. Because the differentiation variable is the source position $\boldsymbol{y}$, the observation position $\boldsymbol{x}$ is constant. In Cartesian components,

$$\nabla_{\boldsymbol{y}} \cdot \boldsymbol{d} = \frac{\partial(y_1 - x_1)}{\partial y_1} + \frac{\partial(y_2 - x_2)}{\partial y_2} + \frac{\partial(y_3 - x_3)}{\partial y_3} = 1 + 1 + 1 = 3. \tag{16}$$

Furthermore,

$$\begin{aligned} \nabla_{\boldsymbol{y}} \left(\frac{1}{l}\right) &= \nabla_{\boldsymbol{y}} \left[(\boldsymbol{d} \cdot \boldsymbol{d})^{-1/2}\right] \\ &= -\frac{1}{2}(\boldsymbol{d} \cdot \boldsymbol{d})^{-3/2} \nabla_{\boldsymbol{y}} (\boldsymbol{d} \cdot \boldsymbol{d}) = -\frac{\boldsymbol{d}}{l^3}. \end{aligned} \tag{17}$$

Substituting Eqs. (16) and (17) into Eq. (15) gives

$$\begin{aligned}\nabla_{\boldsymbol{y}} \cdot \left(\frac{\boldsymbol{d}}{2l}\right) &= \frac{1}{2}\left[\frac{1}{l}\nabla_{\boldsymbol{y}} \cdot \boldsymbol{d} + \boldsymbol{d} \cdot \nabla_{\boldsymbol{y}} \left(\frac{1}{l}\right)\right] \\ &= \frac{1}{2}\left(\frac{3}{l} - \frac{\boldsymbol{d} \cdot \boldsymbol{d}}{l^3}\right) = \frac{1}{l}. \end{aligned} \tag{18}$$

Hence the Newton kernel has the exact divergence representation

$$\boxed{\frac{1}{l} = \nabla_{\boldsymbol{y}} \cdot \boldsymbol{F}_0, \qquad \boldsymbol{y} \neq \boldsymbol{x}.} \tag{19}$$

2.3 Application to an exterior observation point

Assume first that $P \notin \overline{\Omega}$. Then the vector field $\boldsymbol{F}_0$ is continuously differentiable on $\overline{\Omega}$, so all conditions of the divergence theorem are satisfied. Substitute Eq. (19) into the Newton volume potential:

$$\begin{aligned} V &= G\rho \int_\Omega \frac{1}{l}\, \mathrm{d}\Omega \\ &= G\rho \int_\Omega \nabla_{\boldsymbol{y}} \cdot \boldsymbol{F}_0\, \mathrm{d}\Omega. \end{aligned} \tag{20}$$

Applying Eq. (12) to the last integral yields

$$\boxed{V = G\rho \int_{\partial\Omega} \boldsymbol{F}_0 \cdot \boldsymbol{n}\, \mathrm{d}S,} \tag{21}$$

where $\boldsymbol{n}$ is the outward unit normal of the tesseroid. Equation (21) is exact: no quadrature rule, series expansion, or far-field approximation has been introduced.

2.4 Interior observation point

If $P \in \Omega$, the vector field is undefined at the single point $\boldsymbol{y} = \boldsymbol{x}$, so the divergence theorem cannot be applied directly to the unpunctured volume. Let $B_\delta(P)$ be a ball of radius δ centred at P, and define the punctured domain

$$\Omega_\delta = \Omega \setminus B_\delta(P). \tag{22}$$

The divergence theorem is valid on Ω_δ. Its boundary consists of the original boundary $\partial\Omega$ and the small inner sphere $\partial B_\delta(P)$. The outward normal of the punctured domain on the inner sphere points into the removed ball, hence

$$\boldsymbol{n}_\delta = -\frac{\boldsymbol{d}}{l}. \tag{23}$$

Since $\|\boldsymbol{F}_0\| = 1/2$, the small-sphere flux is

$$\int_{\partial B_\delta(P)} \boldsymbol{F}_0 \cdot \boldsymbol{n}_\delta\, \mathrm{d}S = -\frac{1}{2}\int_{\partial B_\delta(P)} 1\, \mathrm{d}S = -2\pi\delta^2 \longrightarrow 0. \tag{24}$$

The removed Newton integral also tends to zero:

$$\int_{B_\delta(P)} \frac{1}{l}\, \mathrm{d}\Omega = 4\pi \int_0^\delta s\, \mathrm{d}s = 2\pi\delta^2 \longrightarrow 0. \tag{25}$$

Taking $\delta \to 0$ proves that Eq. (21) is valid for a strict interior point as well. No Dirac delta term appears here because $\boldsymbol{F}_0$ is bounded; the familiar delta distribution occurs instead for the more singular field $\boldsymbol{d}/l^3$.

2.5 The six oriented boundary faces

The closed boundary consists of two spherical faces and four side faces. Their outward normals and surface elements are

face	outward unit normal	surface element
$r' = r_2$	$+\boldsymbol{e}_{r'}$	$r_2^2 \cos\varphi' \, \mathrm{d}\varphi' \mathrm{d}\lambda'$
$r' = r_1$	$-\boldsymbol{e}_{r'}$	$r_1^2 \cos\varphi' \, \mathrm{d}\varphi' \mathrm{d}\lambda'$
$\varphi' = \varphi_2$	$+\boldsymbol{e}_{\varphi'}$	$r' \cos\varphi_2 \, \mathrm{d}r' \mathrm{d}\lambda'$
$\varphi' = \varphi_1$	$-\boldsymbol{e}_{\varphi'}$	$r' \cos\varphi_1 \, \mathrm{d}r' \mathrm{d}\lambda'$
$\lambda' = \lambda_2$	$+\boldsymbol{e}_{\lambda'}$	$r' \, \mathrm{d}r' \mathrm{d}\varphi'$
$\lambda' = \lambda_1$	$-\boldsymbol{e}_{\lambda'}$	$r' \, \mathrm{d}r' \mathrm{d}\varphi'$

(26)

The signs in this table are exactly the signs represented by $\epsilon_1 = -1$ and $\epsilon_2 = +1$.

3. Analytical radial primitive for the four side faces

The gauge-optimized field constructed below contains one radial primitive on each angular side face. That primitive is analytical.

Assume $r > 0$ and introduce the dimensionless radius

$$R = \frac{r'}{r}, \qquad R_i = \frac{r_i}{r}, \qquad \ell = \frac{l}{r} = \sqrt{R^2 - 2Rc + 1}. \tag{27}$$

On a fixed angular side face, c is independent of r'. The radial factor that appears in the boundary flux is

$$\begin{aligned} r \int_{r_1}^{r_2} \frac{r'}{l} \, \mathrm{d}r' &= r \int_{R_1}^{R_2} \frac{rR}{r\ell} \, r \, \mathrm{d}R \\ &= r^2 \int_{R_1}^{R_2} \frac{R}{\ell} \, \mathrm{d}R. \end{aligned} \tag{28}$$

Define

$$M_1^{(1)}(c) := \int_{R_1}^{R_2} \frac{R}{\ell} \, \mathrm{d}R, \qquad \mathcal{H}(c) := r^2 M_1^{(1)}(c). \tag{29}$$

Equation (28) is therefore

$$r \int_{r_1}^{r_2} \frac{r'}{l} \, \mathrm{d}r' = \mathcal{H}(c). \tag{30}$$

To evaluate $M_1^{(1)}$, set

$$u = R - c, \qquad s = \sqrt{1 - c^2}. \tag{31}$$

Then

$$R = u + c, \qquad \ell = \sqrt{u^2 + s^2}, \qquad \mathrm{d}R = \mathrm{d}u. \tag{32}$$

Hence

$$\begin{aligned}\int \frac{R}{\ell}\,\mathrm{d}R &= \int \frac{u+c}{\sqrt{u^2+s^2}}\,\mathrm{d}u \\ &= \int \frac{u}{\sqrt{u^2+s^2}}\,\mathrm{d}u + c\int \frac{1}{\sqrt{u^2+s^2}}\,\mathrm{d}u \\ &= \sqrt{u^2+s^2} + c\,\operatorname{asinh}\left(\frac{u}{s}\right), \qquad s>0. \end{aligned} \tag{33}$$

Let

$$u_i = R_i - c, \qquad L_i = \sqrt{u_i^2+s^2}, \quad i=1,2. \tag{34}$$

Evaluation of Eq. (33) at the two radial endpoints gives

$$\boxed{M_1^{(1)}(c) = L_2 - L_1 + c\left[\operatorname{asinh}\left(\frac{u_2}{s}\right) - \operatorname{asinh}\left(\frac{u_1}{s}\right)\right].} \tag{35}$$

The collinear cases $s=0$ are obtained by the continuous limit whenever the observation point is not on the corresponding face. In a numerical program, one may also retain the exact integral definition in Eq. (29) or use the cancellation-resistant physical-distance form derived later.

4. Mean-zero spherical correction and the gauge-optimized field

Consider a sphere of radius $a>0$, oriented by the normal $+\boldsymbol{e}_{r'}$. Define

$$l_a(c) = \sqrt{(a-r)^2 + 2ar(1-c)}, \qquad f_a(c) = \frac{a-rc}{l_a(c)}. \tag{36}$$

The radial component of the normalized field in Eq. (14) is $\boldsymbol{F}_0\cdot\boldsymbol{e}_{r'} = f_a(c)/2$. We shall remove its angular variation without changing its three-dimensional divergence.

Define the complete-sphere mean

$$m_a := \frac{1}{2}\int_{-1}^{1} f_a(t)\,\mathrm{d}t. \tag{37}$$

With the substitution $z = l_a(t)$,

$$t = \frac{a^2+r^2-z^2}{2ar}, \qquad \mathrm{d}t = -\frac{z}{ar}\,\mathrm{d}z, \qquad a-rt = \frac{a^2-r^2+z^2}{2a}, \tag{38}$$

and therefore

$$m_a = \frac{1}{4a^2r}\int_{|a-r|}^{a+r}\left(a^2-r^2+z^2\right)\,\mathrm{d}z. \tag{39}$$

It follows that

$$\boxed{m_a = \begin{cases} 1-\dfrac{r^2}{3a^2}, & 0\le r<a, \\ \dfrac{2}{3}, & r=a, \\ \dfrac{2a}{3r}, & r>a. \end{cases}} \tag{40}$$

The angular area of the patch on the unit sphere is

$$\Omega_\Gamma = (\lambda_2-\lambda_1)(\sin\varphi_2 - \sin\varphi_1). \tag{41}$$

The role of this mean will become explicit in the radial flux below.

On the sphere S_a,

$$\nabla_{S_a} c = \frac{1}{a}\left(C_\varphi \boldsymbol{e}_{\varphi'} + C_\lambda \boldsymbol{e}_{\lambda'}\right), \tag{42}$$

$$|\nabla_{S_a} c|^2 = \frac{1-c^2}{a^2}, \qquad \Delta_{S_a} c = -\frac{2c}{a^2}. \tag{43}$$

Introduce a scalar coefficient $\mathcal{Q}_a(c)$ through the mean-zero spherical equation

$$\frac{1}{a^2}\frac{\mathrm{d}}{\mathrm{d}c}\left[(1-c^2)\mathcal{Q}_a(c)\right] = f_a(c) - m_a. \tag{44}$$

Its endpoint-regular solution is

$$\boxed{\mathcal{Q}_a(c) = \frac{a^2}{1-c^2}\int_{-1}^{c} \left[f_a(t) - m_a\right]\,\mathrm{d}t.} \tag{45}$$

Because m_a is the complete-sphere mean,

$$\int_{-1}^{1} \left[f_a(t) - m_a\right]\,\mathrm{d}t = 0, \tag{46}$$

so the apparent singularities at $c = \pm 1$ are removable.

For $r > 0$, define

$$\mathcal{P}_a(c) = -\frac{(a^2 - r^2)l_a(c) + \frac{1}{3}l_a^3(c)}{2a^2 r}. \tag{47}$$

Direct differentiation gives

$$\mathcal{P}_a'(c) = f_a(c), \tag{48}$$

so Eq. (45) becomes

$$\boxed{\mathcal{Q}_a(c) = \frac{a^2}{1-c^2}\left[\mathcal{P}_a(c) - \mathcal{P}_a(-1) - m_a(c+1)\right].} \tag{49}$$

The removable endpoint values are

$$\boxed{\mathcal{Q}_a(-1) = \frac{a^2}{2}(1 - m_a),} \tag{50}$$

$$\boxed{\mathcal{Q}_a(1) = \frac{a^2}{2}\left[m_a - \operatorname{sgn}(a-r)\right], \qquad a \neq r,} \tag{51}$$

and

$$\boxed{\mathcal{Q}_a(1) = \frac{a^2}{3}, \qquad a = r.} \tag{52}$$

These follow directly from Eq. (45) by l'Hopital's rule, using $f_a(-1) = 1$, $f_a(1) = \operatorname{sgn}(a-r)$ for $a \neq r$, and $\lim_{c\to 1} f_a(c) = 0$ for $a = r$.

The regularity needed below is worth recording explicitly. If $a \neq r$, then f_a is smooth on $[-1, 1]$. The integral in Eq. (45) vanishes to first order at both endpoints, and division by $1 - c^2$ therefore defines a smooth extension of $\mathcal{Q}_a$ to the closed interval. If $a = r$, then

$$f_a(c) = \sqrt{\frac{1-c}{2}}, \qquad \mathcal{Q}_a(c) = \frac{2a^3}{3\left(2a + a\sqrt{2(1-c)}\right)}. \tag{53}$$

In this equal-radius case $\mathcal{Q}_a$ is continuous, but its c derivative is singular at $c = 1$. The only coincident point is $\boldsymbol{y} = \boldsymbol{x}$; it is handled by the same punctured-domain limit as in Sec. 2.4. Away from that point, $\mathcal{Q}_a(c)(\widehat{\boldsymbol{x}} \times \widehat{\boldsymbol{y}})$ has a regular extension at $c = \pm 1$ because the cross product vanishes there.

4.1 A divergence-free gauge correction

For $r > 0$, put

$$\widehat{\boldsymbol{x}} = \frac{\boldsymbol{x}}{r}, \qquad a = \|\boldsymbol{y}\|, \qquad \widehat{\boldsymbol{y}} = \frac{\boldsymbol{y}}{a}, \qquad c = \widehat{\boldsymbol{x}} \cdot \widehat{\boldsymbol{y}}.$$

Define the vector potential

$$\boxed{\boldsymbol{A}_g(\boldsymbol{y}; \boldsymbol{x}) = \frac{\mathcal{Q}_a(c)}{2a} (\widehat{\boldsymbol{x}} \times \widehat{\boldsymbol{y}})} \tag{54}$$

and the gauge-optimized divergence field

$$\boxed{\boldsymbol{F}_\star = \boldsymbol{F}_0 + \nabla_{\boldsymbol{y}} \times \boldsymbol{A}_g.} \tag{55}$$

Where the fields are smooth,

$$\nabla_{\boldsymbol{y}} \cdot \boldsymbol{F}_\star = \nabla_{\boldsymbol{y}} \cdot \boldsymbol{F}_0 = \frac{1}{l}, \tag{56}$$

because the divergence of a curl vanishes.

Local behaviour of the gauge field at coincidence. Fix $\boldsymbol{x} \neq \boldsymbol{0}$ and put $\boldsymbol{h} = \boldsymbol{y} - \boldsymbol{x}$, $\delta = \|\boldsymbol{h}\|$. In a sufficiently small ball about $\boldsymbol{x}$, excluding its centre,

$$\|\boldsymbol{A}_g(\boldsymbol{y}; \boldsymbol{x})\| \leq C\delta, \qquad \|\nabla_{\boldsymbol{y}} \times \boldsymbol{A}_g(\boldsymbol{y}; \boldsymbol{x})\| \leq C, \tag{57}$$

where C is independent of the direction of approach. After defining $\boldsymbol{A}_g(\boldsymbol{x}; \boldsymbol{x}) = \boldsymbol{0}$, the potential is continuous and locally bounded. Its classical curl on the punctured ball has a bounded representative, and this representative is also the distributional curl. Consequently

$$\nabla_{\boldsymbol{y}} \cdot (\nabla_{\boldsymbol{y}} \times \boldsymbol{A}_g) = 0 \quad \text{in } \mathcal{D}' \tag{58}$$

across the coincident point.

Justification. Resolve $\boldsymbol{h}$ into its component parallel to $\widehat{\boldsymbol{x}}$ and its orthogonal component $\boldsymbol{h}_\perp$. Uniformly as $\delta \to 0$,

$$a - r = O(\delta), \qquad 1 - c = \frac{\|\boldsymbol{h}_\perp\|^2}{2r^2} + O\left(\frac{\delta^3}{r^3}\right), \qquad \|\widehat{\boldsymbol{x}} \times \widehat{\boldsymbol{y}}\| = O(\delta/r). \tag{59}$$

Substitution of $d_a^2 = (a - r)^2 + 2ar(1 - c)$ into the three rational branches in Eq. (89) shows that $\mathcal{Q}_a(c)$ is uniformly bounded in this joint limit. This proves the first estimate in Eq. (57). For the curl, its radial component is $-[f_a(c) - m_a]/2$ by Eq. (65) and is bounded because $|f_a| \leq 1$ and m_a is bounded. Differentiating the same rational branches with d_a as the local variable gives

$$\frac{\sin\psi}{a} \left| \frac{\partial \mathcal{Q}_a}{\partial a} \right| \leq C \frac{r \sin\psi}{d_a} \leq C; \tag{60}$$

Eq. (66) proves the second estimate. The rational expressions are smooth away from $\delta = 0$. To identify their classical curl with the distributional curl, integrate by parts on a punctured ball. The additional boundary term is bounded by

$$C \sup |\boldsymbol{\phi}| \, \delta^2 \sup_{\partial B_\delta} |\boldsymbol{A}_g| = O(\delta^3) \tag{61}$$

for every test field $\boldsymbol{\phi}$, and therefore vanishes. No point-supported curl is created. Since $\boldsymbol{A}_g \in L^1_{\mathrm{loc}}$, distributional partial derivatives commute, which proves Eq. (58). Notice that the argument deliberately does not infer $W^{1,\infty}_{\mathrm{loc}}$ merely from bounds on $\boldsymbol{A}_g$ and its curl; that stronger assertion would require separate estimates for every first derivative and is not needed here.

If $\boldsymbol{x} \in \Omega$, apply the classical divergence theorem first on the punctured domain Ω_δ. The local estimate in Eq. (57) gives

$$\left| \int_{\partial B_\delta(\boldsymbol{x})} (\nabla \times \boldsymbol{A}_g) \cdot \boldsymbol{n} \, \mathrm{d}S \right| \leq 4\pi C \delta^2 \longrightarrow 0. \tag{62}$$

The $\boldsymbol{F}_0$ contribution tends to zero by Eq. (24); hence the limit and the use of Eq. (56) are legitimate.

The artificial interface $a = r$. Fix a closed angular set on which $c \leq 1 - \eta$ for some $\eta > 0$. As a function of the source radius a, m_a is C^1 at $a = r$ but is not generally C^2 there. The two branches of $\mathcal{Q}_a(c)$ and their first a derivatives have the same trace at $a = r$. Hence $\boldsymbol{A}_g$ is continuous and piecewise C^1 across that sphere. If a source domain is split at $a = r$, the two interface fluxes of $\boldsymbol{F}_\star$ cancel exactly.

Justification. The formulas in Eq. (40) give

$$m_r = \frac{2}{3}, \qquad \partial_a m_a|_{a=r^-} = \partial_a m_a|_{a=r^+} = \frac{2}{3r}, \tag{63}$$

whereas the second derivatives have different traces. For $c \leq 1 - \eta$, d_a is bounded away from zero, so substitution in Eq. (89) and one differentiation in a prove the matching traces of $\mathcal{Q}_a$. Equivalently, the same conclusion follows from the defining integral Eq. (45). Finally, Eq. (65) gives the normal trace

$$\boldsymbol{F}_\star \cdot \boldsymbol{e}_{r'} = \frac{m_a}{2}. \tag{64}$$

The two subdomains assign opposite normals to their common sphere, while m_a has a common trace; their interface integrals are therefore equal and opposite. No delta contribution is created in Eq. (58).

To evaluate the curl explicitly, take $\widehat{\boldsymbol{x}}$ as polar axis, let $c = \cos\psi$, and write $\widehat{\boldsymbol{x}} \times \widehat{\boldsymbol{y}} = \sin\psi \, \boldsymbol{e}_\alpha$. Thus $(A_g)_\alpha = \mathcal{Q}_a(c) \sin\psi/(2a)$. The spherical curl formulas and Eq. (44) give

$$(\nabla_{\boldsymbol{y}} \times \boldsymbol{A}_g)_a = \frac{1}{a \sin\psi} \frac{\partial}{\partial \psi} \left(\sin\psi (A_g)_\alpha \right) = -\frac{f_a(c) - m_a}{2}, \tag{65}$$

$$(\nabla_{\boldsymbol{y}} \times \boldsymbol{A}_g)_\psi = -\frac{1}{a} \frac{\partial}{\partial a} \left(a (A_g)_\alpha \right) = -\frac{\sin\psi}{2a} \frac{\partial \mathcal{Q}_a(c)}{\partial a}, \tag{66}$$

where the source-radius derivative holds c fixed. Since

$$(\boldsymbol{F}_0)_a = \frac{f_a(c)}{2}, \qquad (\boldsymbol{F}_0)_\psi = \frac{r \sin\psi}{2 l_a(c)}, \qquad \nabla_{S_a} c = -\frac{\sin\psi}{a} \boldsymbol{e}_\psi,$$

the complete field has the coordinate-free form

$$\boxed{\boldsymbol{F}_\star = \frac{m_a}{2}\boldsymbol{e}_{r'} + \frac{1}{2}\left[\frac{\partial \mathcal{Q}_a(c)}{\partial a} - \frac{ar}{l_a(c)}\right]\nabla_{S_a} c.} \tag{67}$$

The nonuniform radial flux has therefore been converted into a constant radial flux, while all angular information has been transferred to the tangential part.

4.2 Flux through the six faces

On the two spherical faces the tangential part does not contribute. Their combined outward flux is

$$\frac{1}{2}\left(r_2^2 m_{r_2} - r_1^2 m_{r_1}\right)\Omega_\Gamma = \frac{C_0}{2}. \tag{68}$$

On $\varphi' = \varphi_k$, use $\nabla_{S_a} c \cdot \boldsymbol{e}_{\varphi'} = C_\varphi/a$ and $\mathrm{d}S = a\cos\varphi_k\,\mathrm{d}a\,\mathrm{d}\lambda'$. Integration in a gives

$$\begin{aligned}\int_{\varphi'=\varphi_k} \boldsymbol{F}_\star \cdot \boldsymbol{n}\,\mathrm{d}S &= \frac{\epsilon_k}{2}\cos\varphi_k \int_{\lambda_1}^{\lambda_2} C_\varphi(\varphi_k, \lambda') \\ &\quad \times \left[\mathcal{Q}_{r_2}(c) - \mathcal{Q}_{r_1}(c) - r\int_{r_1}^{r_2} \frac{a}{l_a(c)}\,\mathrm{d}a\right]\mathrm{d}\lambda' \\ &= \frac{\epsilon_k}{2}\cos\varphi_k \int_{\lambda_1}^{\lambda_2} C_\varphi(\varphi_k, \lambda')\mathcal{W}(c)\,\mathrm{d}\lambda'. \end{aligned} \tag{69}$$

Similarly, on $\lambda' = \lambda_k$,

$$\int_{\lambda'=\lambda_k} \boldsymbol{F}_\star \cdot \boldsymbol{n}\,\mathrm{d}S = \frac{\epsilon_k}{2}\int_{\varphi_1}^{\varphi_2} C_\lambda(\varphi', \lambda_k)\mathcal{W}(c)\,\mathrm{d}\varphi'. \tag{70}$$

Thus one application of the three-dimensional divergence theorem produces the explicit spherical term and the four one-dimensional side-face integrals directly; no second surface-divergence theorem is needed.

4.3 Scope of the gauge normalization

The construction has a useful restricted uniqueness property. On every compact radial interval in $(0, \infty)$, consider axisymmetric poloidal fields

$$\boldsymbol{F} = U(a)\boldsymbol{e}_{r'} + Z(a, c)\nabla_{S_a} c,$$

with $U \in W^{1,1}_{\mathrm{loc}}$, $Z(a, \cdot) \in W^{1,1}(-1, 1)$ for almost every a, finite endpoint traces of Z, and locally integrable divergence. Prescribe $\nabla \cdot \boldsymbol{F} = 1/l_a(c)$ and exclude an artificial $a^{-2}\boldsymbol{e}_{r'}$ point-source field at the origin. Integrating $\nabla \cdot \boldsymbol{F} = 1/l_a(c)$ over $c \in [-1, 1]$ gives

$$2\frac{\mathrm{d}}{\mathrm{d}a}\left(a^2 U\right) = a^2 \int_{-1}^{1} \frac{\mathrm{d}c}{l_a(c)}.$$

Using the definition of m_a shows $U = m_a/2 + C/a^2$; the no-point-source condition sets $C = 0$, and endpoint regularity then fixes the poloidal coefficient Z as in Eq. (67). More explicitly, the difference of two admissible fields has $\partial_c[(1 - c^2)\Delta Z] = 0$; finite endpoint traces force the integration constant to vanish. Thus uniqueness holds in precisely this weak axisymmetric poloidal class with the stated endpoint and origin conditions. It is not an unrestricted vector-field uniqueness theorem: regular toroidal curls and other gauge transformations lie outside the class

and may still be added. Such additions redistribute side-face fluxes but do not, in general, remove their total for a finite nonspherical angular patch. Accordingly, the four line integrals are retained as the generic boundary data rather than claimed to be absolutely irreducible.

5. Final one-dimensional boundary representation

Define the combined boundary kernel

$$\boxed{\mathcal{W}(c) := \mathcal{Q}_{r_2}(c) - \mathcal{Q}_{r_1}(c) - \mathcal{H}(c).} \tag{71}$$

The three terms in Eq. (71) all have dimensions of length squared.

For $r > 0$, constant ρ, and $P \notin \partial\Omega$, combining the two spherical-face fluxes and the four lateral-face fluxes gives the exact potential

$$\boxed{V = \frac{G\rho}{2}\left[C_0 + I_{\varphi_1} + I_{\varphi_2} + I_{\lambda_1} + I_{\lambda_2}\right],} \tag{72}$$

where

$$\boxed{C_0 = \left(r_2^2 m_{r_2} - r_1^2 m_{r_1}\right)(\lambda_2 - \lambda_1)(\sin\varphi_2 - \sin\varphi_1),} \tag{73}$$

$$\boxed{I_{\varphi_k} = \epsilon_k \cos\varphi_k \int_{\lambda_1}^{\lambda_2} C_\varphi(\varphi_k, \lambda')\mathcal{W}\big(c(\varphi_k, \lambda')\big)\, \mathrm{d}\lambda', \quad k = 1, 2,} \tag{74}$$

and

$$\boxed{I_{\lambda_k} = \epsilon_k \int_{\varphi_1}^{\varphi_2} C_\lambda(\varphi', \lambda_k)\mathcal{W}\big(c(\varphi', \lambda_k)\big)\, \mathrm{d}\varphi', \quad k = 1, 2.} \tag{75}$$

Derivation. Replace $\boldsymbol{F}_0$ in Eq. (21) by $\boldsymbol{F}_\star$. This leaves the volume divergence unchanged by Eq. (56). The two spherical-face fluxes are Eq. (68), and the four side-face fluxes are Eqs. (69) and (70). Multiplication by $G\rho$ gives Eqs. (72)–(75).

The result consists of one explicit term and four one-dimensional integrals. The gauge field makes the spherical flux constant and transfers its mean-zero part to the side faces. Radial integration on each side then produces the single combined kernel $\mathcal{W}$; no separate spherical-face quadrature, surface-divergence theorem, or radial-edge integration is required.

5.1. Cancellation-resistant form of the gauge coefficient

The representation in Eq. (49) is exact, but it is not the best form for numerical evaluation. Its numerator and denominator both vanish as $c \to \pm 1$, so a direct floating-point implementation may suffer from severe cancellation. This subsection removes the factor $1 - c^2$ analytically. The resulting formulas are shorter, contain no removable $0/0$ form, and are easier to interpret physically.

For a spherical face of radius a, introduce the physical source–observation distance

$$\boxed{d_a = d_a(c) := \sqrt{(a - r)^2 + 2ar(1 - c)}.} \tag{76}$$

This is algebraically equal to $\sqrt{a^2 + r^2 - 2arc}$, but the displayed form is the required floating-point evaluation: it does not subtract two nearly equal squared radii. When the source and

observation directions are available as unit vectors, compute

$$s := 1 - c = \frac{1}{2}\|\widehat{\boldsymbol{y}} - \widehat{\boldsymbol{x}}\|^2, \qquad d_a = \mathrm{hypot}\Big(a - r, \sqrt{2ar\,s}\Big)\,. \tag{77}$$

Equivalently, with the stable angle $\psi = \mathrm{atan2}(\|\widehat{\boldsymbol{x}} \times \widehat{\boldsymbol{y}}\|, c)$, the second argument of hypot is $2\sqrt{ar}\sin(\psi/2)$. Thus Eq. (76) removes radial cancellation, while Eq. (77) also prevents the dot product from rounding a very small nonzero angular separation to $c = 1$. Equation (76) can be inverted as

$$c = \frac{a^2 + r^2 - d_a^2}{2ar}. \tag{78}$$

Moreover,

$$\begin{aligned} 1 - c^2 &= \frac{4a^2r^2 - (a^2 + r^2 - d_a^2)^2}{4a^2r^2} \\ &= \frac{\left[(a+r)^2 - d_a^2\right]\left[d_a^2 - (a-r)^2\right]}{4a^2r^2} \\ &= \frac{(a + r - d_a)(a + r + d_a)(d_a - a + r)(d_a + a - r)}{4a^2r^2}. \end{aligned} \tag{79}$$

This factorization is the key to the cancellation below.

For convenience, define the numerator of Eq. (49) by

$$N_a(c) := a^2\left[\mathcal{P}_a(c) - \mathcal{P}_a(-1) - m_a(c+1)\right], \qquad \mathcal{Q}_a(c) = \frac{N_a(c)}{1 - c^2}. \tag{80}$$

Because m_a is piecewise in the relative positions of r and a, the factorization of N_a is also most transparently derived in three cases.

Case 1: the observation point is inside the sphere, $0 < r < a$. In this case

$$m_a = 1 - \frac{r^2}{3a^2}. \tag{81}$$

Substitution of Eqs. (47), (78), and the above m_a into Eq. (80), followed only by collection of polynomial factors, gives

$$N_a(c) = \frac{(d_a + r - a)(a + r - d_a)\left(r^2 - a^2 + ad_a\right)}{6ar}. \tag{82}$$

The factors $(d_a + r - a)$ and $(a + r - d_a)$ also occur in Eq. (79). Cancelling these common factors yields

$$\boxed{\mathcal{Q}_a(c) = \frac{2ar\left(r^2 - a^2 + ad_a\right)}{3\left[(a + d_a)^2 - r^2\right]}, \qquad 0 < r < a.} \tag{83}$$

The denominator is strictly positive because $a + d_a > r$.

Case 2: the observation point is outside the sphere, $r > a$. Now

$$m_a = \frac{2a}{3r}. \tag{84}$$

The corresponding factorization of Eq. (80) is

$$N_a(c) = \frac{(d_a + a - r)(a + r - d_a)\left[rd_a + 2(r^2 - a^2)\right]}{6r^2}. \tag{85}$$

After cancelling the common factors with Eq. (79), one obtains

$$\boxed{\mathcal{Q}_a(c) = \frac{2a^2\left[rd_a + 2(r^2-a^2)\right]}{3\left[(r+d_a)^2 - a^2\right]}, \qquad r > a.} \tag{86}$$

Again, the denominator is strictly positive because $r + d_a > a$.

Case 3: the observation point lies on the complete sphere, $r = a$. Here $m_a = 2/3$ and

$$d_a = a\sqrt{2(1-c)}. \tag{87}$$

Direct substitution into Eq. (49), or equivalently the common limit of Eqs. (83) and (86), gives

$$\boxed{\mathcal{Q}_a(c) = \frac{2a^3}{3(2a+d_a)}, \qquad r = a.} \tag{88}$$

This formula is finite for all $-1 \le c \le 1$. Notice that $r = a$ does not necessarily mean that P lies on the tesseroid boundary: the angular coordinates of P may be outside the spherical patch.

Combining the three cases gives the cancellation-resistant expression

$$\boxed{\mathcal{Q}_a(c) = \begin{cases} \dfrac{2ar\left(r^2 - a^2 + ad_a\right)}{3\left[(a+d_a)^2 - r^2\right]}, & 0 < r < a, \\[2ex] \dfrac{2a^3}{3(2a+d_a)}, & r = a, \\[2ex] \dfrac{2a^2\left[rd_a + 2(r^2-a^2)\right]}{3\left[(r+d_a)^2 - a^2\right]}, & r > a. \end{cases}} \tag{89}$$

Equation (89) is algebraically identical to Eq. (49); it merely cancels the factors responsible for the removable angular singularities. It is useful analytically, but the differences $r^2 - a^2$ and $r^2 - a^2 + ad_a$ can still lose digits when $a \simeq r$. A floating-point implementation must also remove this radial cancellation. Put $s = 1 - c$. For $r \neq a$, let $\delta = |a - r|$ and use $d_a - \delta = 2ars/(d_a + \delta)$. The fully stabilized branches are

$$\boxed{\mathcal{Q}_a(c) = \begin{cases} \dfrac{2ar^2}{3(d_a+\delta)(a+d_a+r)}\left[\dfrac{2a^2 s}{d_a+\delta} - \delta\right], & r < a, \quad \delta = a - r, \\[2ex] \dfrac{2a^3}{3(2a+d_a)}, & r = a, \\[2ex] \dfrac{2a^2\left[rd_a + 2\delta(r+a)\right]}{3(d_a+\delta)(r+d_a+a)}, & r > a, \quad \delta = r - a. \end{cases}} \tag{90}$$

These expressions contain neither a difference of squared radii nor the difference of two $O(r^2)$ quantities. Equation (90), together with the chord-based computation of s in Eq. (77), is the required numerical form.

For a tesseroid, the branch used for each radial boundary is immediately identified from the observation radius:

observation radius	$\mathcal{Q}_{r_1}$	$\mathcal{Q}_{r_2}$
$0 < r < r_1$	$r < r_1$ branch	$r < r_2$ branch
$r_1 < r < r_2$	$r > r_1$ branch	$r < r_2$ branch
$r > r_2$	$r > r_1$ branch	$r > r_2$ branch

(91)

The equal-radius formulas are used at $r = r_1$ or $r = r_2$, subject to the boundary restriction stated below Eq. (7).

5.2. Physical-distance form of the radial side-face kernel

The normalized expression in Eq. (35) can also be written directly in physical distances. Define

$$d_i(c) := \sqrt{(r_i - r)^2 + 2r_i r(1 - c)}, \qquad i = 1, 2. \tag{92}$$

Because $L_i = d_i/r$ and $u_i = (r_i - rc)/r$, the first part of $r^2 M_1^{(1)}$ is

$$r^2(L_2 - L_1) = r(d_2 - d_1). \tag{93}$$

For $-1 < c < 1$,

$$\operatorname{asinh}\left(\frac{u_i}{\sqrt{1-c^2}}\right) = \ln\left[\frac{r_i - rc + d_i}{r\sqrt{1-c^2}}\right]. \tag{94}$$

The common denominator in Eq. (94) cancels when the two radial endpoints are subtracted. Consequently,

$$\boxed{\mathcal{H}(c) = r\,[d_2(c) - d_1(c)] + r^2 c \ln\left[\frac{r_2 - rc + d_2(c)}{r_1 - rc + d_1(c)}\right].} \tag{95}$$

The right-hand side extends by continuity to the collinear cases whenever $P \notin \partial\Omega$.

The displayed quotient is an analytical identity, not a safe instruction to form both of its numerator and denominator. Define

$$u_i := r_i - rc, \qquad z_i := u_i + d_i. \tag{96}$$

Since $d_i^2 - u_i^2 = r^2(1 - c^2)$, evaluate each logarithm by

$$\boxed{\ell_i := \ln z_i = \begin{cases} \ln(d_i + u_i), & u_i \geq 0, \\ 2\ln r + \ln(1-c) + \ln(1+c) - \ln(d_i - u_i), & u_i < 0, \end{cases}} \tag{97}$$

and compute

$$\boxed{\mathcal{H}(c) = r(d_2 - d_1) + r^2 c(\ell_2 - \ell_1).} \tag{98}$$

If both $u_1, u_2 < 0$, the common singular factor is cancelled before taking a logarithm:

$$\ell_2 - \ell_1 = \ln\frac{d_1 - u_1}{d_2 - u_2}. \tag{99}$$

This rule also covers a mixed-sign radial interval, which cannot be handled by a single direct quotient without cancelling in its negative-u_i endpoint.

For an explicit collinear evaluation, write $\Delta r = r_2 - r_1$. Directly integrating $rr'/|r' - r|$ gives

$$\boxed{\mathcal{H}(1) = \begin{cases} r\left[\Delta r + r\ln\dfrac{r_2 - r}{r_1 - r}\right], & 0 < r < r_1, \\ r\left[r\ln\dfrac{r - r_1}{r - r_2} - \Delta r\right], & r > r_2, \end{cases}} \tag{100}$$

and

$$\boxed{\mathcal{H}(-1) = r\left[\Delta r - r\ln\frac{r+r_2}{r+r_1}\right].} \tag{101}$$

The missing case $r_1 \le r \le r_2$ in Eq. (100) has a logarithmic singularity and corresponds to $P \in \partial\Omega$ when the coincident direction lies on an angular face. Derivatives of the collinear branch are obtained from Eqs. (213)–(216) using the finite-sum moments (208)–(210).

Using Eqs. (89) and (95), the common boundary kernel becomes

$$\boxed{\begin{aligned}\mathcal{W}(c) = q_2(c) - q_1(c) - r\left[d_2(c) - d_1(c)\right] \\ - r^2 c\ln\left[\frac{r_2 - rc + d_2(c)}{r_1 - rc + d_1(c)}\right],\end{aligned}} \tag{102}$$

where

$$q_i(c) := \mathcal{Q}_{r_i}(c), \qquad i = 1, 2, \tag{103}$$

and each q_i is evaluated from the appropriate branch of Eq. (89). Thus every evaluation of $\mathcal{W}$ requires only square roots, rational operations, and one logarithm. The modular form in Eq. (102) is considerably safer than a single fully expanded expression because it preserves the cancellations within each analytically regular component.

5.3. Pairing the two latitude boundaries

The signs $\epsilon_1 = -1$ and $\epsilon_2 = +1$ express the opposite outward co-normal orientations of the lower and upper latitude boundaries. They allow the two latitude-edge contributions to be written as one parameter integral.

For fixed $\varphi' = \varphi_k$, define

$$c_{\varphi,k}(\lambda') := \sin\varphi\sin\varphi_k + \cos\varphi\cos\varphi_k\cos(\lambda' - \lambda), \qquad k = 1, 2. \tag{104}$$

The following identity removes the separate factor $\cos\varphi_k C_\varphi$:

$$\begin{aligned}\sin\varphi - \sin\varphi_k c_{\varphi,k} &= \sin\varphi - \sin\varphi_k\left[\sin\varphi\sin\varphi_k + \cos\varphi\cos\varphi_k\cos(\lambda' - \lambda)\right] \\ &= \sin\varphi\cos^2\varphi_k - \cos\varphi\sin\varphi_k\cos\varphi_k\cos(\lambda' - \lambda) \\ &= \cos\varphi_k\left[\sin\varphi\cos\varphi_k - \cos\varphi\sin\varphi_k\cos(\lambda' - \lambda)\right] \\ &= \cos\varphi_k C_\varphi(\varphi_k, \lambda').\end{aligned} \tag{105}$$

Substituting Eq. (105) into Eq. (74), and then using $\epsilon_1 = -1$ and $\epsilon_2 = +1$, gives

$$\boxed{\begin{aligned}I_\varphi &:= I_{\varphi_1} + I_{\varphi_2} \\ &= \int_{\lambda_1}^{\lambda_2}\Big\{\left[\sin\varphi - \sin\varphi_2 c_{\varphi,2}(\lambda')\right]\mathcal{W}\big(c_{\varphi,2}(\lambda')\big) \\ &\qquad - \left[\sin\varphi - \sin\varphi_1 c_{\varphi,1}(\lambda')\right]\mathcal{W}\big(c_{\varphi,1}(\lambda')\big)\Big\}\,\mathrm{d}\lambda'.\end{aligned}} \tag{106}$$

Equation (106) is an oriented upper-minus-lower boundary difference. No approximation has been introduced.

5.4. Pairing the two longitude boundaries

For fixed $\lambda' = \lambda_k$, define

$$c_{\lambda,k}(\varphi') := \sin\varphi \sin\varphi' + \cos\varphi \cos\varphi' \cos(\lambda_k - \lambda), \qquad k = 1, 2. \tag{107}$$

On this boundary,

$$C_\lambda(\varphi', \lambda_k) = -\cos\varphi \sin(\lambda_k - \lambda), \tag{108}$$

which is independent of the integration variable φ'. Therefore Eq. (75) gives

$$\boxed{\begin{aligned} I_\lambda &:= I_{\lambda_1} + I_{\lambda_2} \\ &= -\cos\varphi \int_{\varphi_1}^{\varphi_2} \Big\{ \sin(\lambda_2 - \lambda) \mathcal{W}\big(c_{\lambda,2}(\varphi')\big) \\ &\qquad\qquad - \sin(\lambda_1 - \lambda) \mathcal{W}\big(c_{\lambda,1}(\varphi')\big) \Big\} \, \mathrm{d}\varphi'. \end{aligned}} \tag{109}$$

This is the oriented east-minus-west boundary difference.

Two paired one-dimensional parameter integrals. Under the conditions stated for Eq. (72), the potential can equivalently be written as

$$\boxed{V = \frac{G\rho}{2} \left[C_0 + I_\varphi + I_\lambda \right],} \tag{110}$$

where C_0 is given by Eq. (73), I_φ by Eq. (106), and I_λ by Eq. (109).

Equation (110) is the final potential representation adopted in this derivation. No further reduction to elliptic or other special functions is assumed.

The phrase "two paired integrals" concerns the organization of the formula, not a factor-of-two reduction in the number of kernel evaluations. At a given quadrature node of Eq. (106), the kernel must still be evaluated at both $c_{\varphi,1}$ and $c_{\varphi,2}$; similarly, Eq. (109) requires both longitude boundaries. The paired form nevertheless has three practical advantages: both boundaries use the same quadrature nodes, common trigonometric quantities can be reused, and the formula is already expressed as an oriented boundary difference. The last property is particularly important when many neighboring tesseroids are assembled, because identical shared boundaries with identical physical properties cancel exactly.

5.5. Cancellation-resistant reformulation for small tesseroids

Equations (106) and (109) are exact and already use the correct oriented differences of opposite boundaries. When the angular widths or the radial thickness of the tesseroid become very small, however, the two terms inside either oriented difference can be nearly equal. Direct floating-point subtraction can then lose significant digits. This subsection rewrites the same formulas so that the small geometric widths appear as explicit multiplicative factors. No approximation is introduced in the identities themselves.

5.5.1. Divided differences and the source of the cancellation

For a differentiable scalar function f, define its first divided difference by

$$f^{[1]}(x_-, x_+) := \begin{cases} \dfrac{f(x_+) - f(x_-)}{x_+ - x_-}, & x_+ \neq x_-, \\ f'(x_-), & x_+ = x_-. \end{cases} \tag{111}$$

The second branch is the continuous limit of the first. The identity

$$f(x_+) - f(x_-) = (x_+ - x_-) f^{[1]}(x_-, x_+) \tag{112}$$

extracts the small separation explicitly.

The notation in Eq. (111) is mathematical; it does not prescribe evaluating the numerator by direct subtraction when x_+ and x_- are very close. Let

$$x_m = \frac{x_+ + x_-}{2}, \qquad h_x = \frac{x_+ - x_-}{2}. \tag{113}$$

A symmetric Taylor expansion gives

$$\boxed{f^{[1]}(x_-, x_+) = f'(x_m) + \frac{h_x^2}{3!} f^{(3)}(x_m) + \frac{h_x^4}{5!} f^{(5)}(x_m) + \cdots .} \tag{114}$$

The series can be stated with a rigorous finite remainder. If $f \in C^{2N+3}([x_m - h_x, x_m + h_x])$, Taylor's theorem applied symmetrically gives

$$f^{[1]}(x_m - h_x, x_m + h_x) = \sum_{j=0}^{N} \frac{h_x^{2j}}{(2j+1)!} f^{(2j+1)}(x_m) + R_{2N+2}, \tag{115}$$

where

$$|R_{2N+2}| \leq \frac{|h_x|^{2N+2}}{(2N+3)!} \max_{|t - x_m| \leq |h_x|} |f^{(2N+3)}(t)|. \tag{116}$$

In particular, the extended divided difference is continuous on the diagonal; if $f \in C^{k+1}$, then $(x_-, x_+) \mapsto f^{[1]}(x_-, x_+)$ is C^k. Thus a robust implementation may use the ordinary quotient for separated arguments and Eq. (115), or an equivalent automatically differentiated local expansion, near the diagonal. A numerical switching threshold is machine dependent and is not part of the mathematical representation.

For reproducibility, the reference IEEE 754 double-precision realization used with this paper fixes the following algorithm. With $S_x = \max(1, |x_-|, |x_+|)$, use the direct quotient when $|x_+ - x_-| > 10^{-4} S_x$; otherwise use the centered odd series through order nine,

$$f^{[1]} = f'(x_m) + \frac{h_x^2}{3!} f^{(3)}(x_m) + \frac{h_x^4}{5!} f^{(5)}(x_m) + \frac{h_x^6}{7!} f^{(7)}(x_m) + \frac{h_x^8}{9!} f^{(9)}(x_m). \tag{117}$$

After the fifth-order term, the sum may stop early only when every requested Taylor-jet component of the newest term is at most $128 \epsilon_{\text{mach}} \max(1, |\text{partial sum}|)$. If a certified bound for $f^{(11)}$ is available, the truncation error is additionally bounded by Eq. (116) with $N = 4$; otherwise the calculation is flagged as a floating-point algorithm rather than a rigorous interval enclosure. For the simultaneous thin-radius/thin-angle difference, the bivariate series through cubic order in each variable is used only when $|\Delta r| \leq 10^{-8} S_r$ and $|\Delta c| \leq 10^{-6} S_c$; outside that small box the two one-dimensional ninth-order rules are applied in sequence. These constants are implementation choices, not new mathematical assumptions.

A second elementary identity will be used repeatedly. For four scalar quantities $A_\pm$ and $W_\pm$, define

$$\overline{A} = \frac{A_+ + A_-}{2}, \qquad \Delta A = A_+ - A_-, \qquad \overline{W} = \frac{W_+ + W_-}{2}, \qquad \Delta W = W_+ - W_-. \tag{118}$$

Direct expansion proves

$$\boxed{A_+W_+ - A_-W_- = \overline{A}\,\Delta W + \overline{W}\,\Delta A.} \tag{119}$$

The right-hand side contains averages multiplied by small differences instead of a direct subtraction of two nearly equal products.

Introduce the cell widths, midpoints, and half-widths

$$\Delta r = r_2 - r_1, \qquad r_c = \frac{r_1 + r_2}{2}, \qquad h_r = \frac{\Delta r}{2}, \tag{120}$$

$$\Delta\varphi = \varphi_2 - \varphi_1, \qquad \varphi_c = \frac{\varphi_1 + \varphi_2}{2}, \qquad h_\varphi = \frac{\Delta\varphi}{2}, \tag{121}$$

$$\Delta\lambda = \lambda_2 - \lambda_1, \qquad \lambda_c = \frac{\lambda_1 + \lambda_2}{2}, \qquad h_\lambda = \frac{\Delta\lambda}{2}. \tag{122}$$

Define the entire function

$$\operatorname{sinc} x := \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases} \tag{123}$$

For small x, it should be evaluated from

$$\operatorname{sinc} x = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \cdots, \tag{124}$$

rather than by dividing two small floating-point numbers.

5.5.2. Stable extraction of the radial thickness

The common kernel in Eq. (102) vanishes when $r_2 = r_1$. This radial smallness can also be extracted exactly. For an arbitrary radial boundary $a > 0$, define

$$d_a(c) = \sqrt{(a - r)^2 + 2ar(1 - c)}. \tag{125}$$

On each connected radial interval on which the observation point is not crossed, introduce a primitive

$$\mathcal{S}(a; r, c) := r\, d_a(c) + r^2 c \ln\left(\frac{a - rc + d_a(c)}{L_0}\right), \tag{126}$$

where $L_0 > 0$ is an arbitrary reference length. The choice of L_0 cancels from every endpoint difference. Equation (95) is equivalently

$$\mathcal{H}(c) = \mathcal{S}(r_2; r, c) - \mathcal{S}(r_1; r, c). \tag{127}$$

At a collinear limiting configuration, Eqs. (126) and (127) are interpreted through the continuous branch already defined by the original radial integral.

Define

$$\mathcal{U}(a; r, c) := \mathcal{Q}_a(r, c) - \mathcal{S}(a; r, c). \tag{128}$$

Then

$$\mathcal{W}(r,c) = \mathcal{U}(r_2;r,c) - \mathcal{U}(r_1;r,c). \tag{129}$$

Consequently,

$$\boxed{\mathcal{W}(r,c) = \Delta r\, \widehat{\mathcal{W}}(r,c), \qquad \widehat{\mathcal{W}}(r,c) := \mathcal{U}^{[1]}(r_1,r_2;r,c).} \tag{130}$$

Here the divided difference is taken with respect to the first, source-radius argument of $\mathcal{U}$; r and c are held fixed. For a very thin radial cell, $\widehat{\mathcal{W}}$ is evaluated by the centered series

$$\widehat{\mathcal{W}}(r,c) = \mathcal{U}_a(r_c;r,c) + \frac{h_r^2}{3!}\mathcal{U}_{aaa}(r_c;r,c) + \frac{h_r^4}{5!}\mathcal{U}_{aaaaa}(r_c;r,c) + \cdots . \tag{131}$$

If the interval $[r_1,r_2]$ straddles $a=r$, the two analytical branches are combined by splitting the endpoint difference at $a=r$ before applying a local series. This avoids differentiating a piecewise expression across its branch point.

More precisely, on every compact interval $K \subset (0,r)$ or $K \subset (r,\infty)$ for which the source and observation directions do not coincide at zero distance, $a \mapsto \mathcal{U}(a;r,c)$ is smooth. Therefore Eq. (115) applies to $\widehat{\mathcal{W}}$. If $r_1 < r < r_2$, the exact identity

$$\mathcal{W} = \big[\mathcal{U}(r_2;r,c) - \mathcal{U}(r;r,c)\big] + \big[\mathcal{U}(r;r,c) - \mathcal{U}(r_1;r,c)\big] \tag{132}$$

is used before either difference is expanded. The equal-radius value in Eq. (88) supplies the common endpoint value. No Taylor expansion is taken across the change of analytical branch. Equivalently, the executable divided difference is the weighted formula

$$\boxed{\mathcal{U}^{[1]}(r_1,r_2;r,c) = \frac{r-r_1}{r_2-r_1}\,\mathcal{U}_-^{[1]}(r_1,r;r,c) + \frac{r_2-r}{r_2-r_1}\,\mathcal{U}_+^{[1]}(r,r_2;r,c),} \tag{133}$$

where the subscripts select the $a<r$ and $a>r$ analytic branches, respectively. Each one-sided divided difference is evaluated by Eq. (117); zero-length pieces are omitted.

The exact factorization in Eq. (130) is not required for moderately thick cells. It is a stable evaluation route for the thin-cell regime, where direct computation of q_2-q_1, d_2-d_1, and the logarithmic ratio can otherwise suffer cancellation.

5.5.3. Cancellation-resistant latitude-boundary difference

For fixed λ', regard the source latitude as a variable α and define

$$c(\alpha) = \sin\varphi\sin\alpha + \cos\varphi\cos\alpha\cos(\lambda'-\lambda), \tag{134}$$

$$A(\alpha) := \sin\varphi - \sin\alpha\, c(\alpha) = \cos\alpha\, C_\varphi(\alpha,\lambda'). \tag{135}$$

Set

$$c_\pm = c(\varphi_c \pm h_\varphi), \qquad A_\pm = A(\varphi_c \pm h_\varphi), \qquad \widehat{W}_\pm = \widehat{\mathcal{W}}(r,c_\pm). \tag{136}$$

The plus sign denotes the upper latitude boundary φ_2, and the minus sign denotes the lower boundary φ_1.

First, use the sum-to-product identities for sine and cosine. Since $c_+ - c_-$ is the difference of Eq. (134) at two points symmetric about φ_c,

$$\begin{aligned} c_+ - c_- &= 2\sin h_\varphi \left[\sin\varphi\cos\varphi_c - \cos\varphi\sin\varphi_c\cos(\lambda'-\lambda)\right] \\ &= \Delta\varphi\ \mathrm{sinc}(h_\varphi) C_\varphi(\varphi_c,\lambda'). \end{aligned} \tag{137}$$

Next, differentiate Eq. (135) with respect to α:

$$A'(\alpha) = -\cos\alpha\, c(\alpha) - \sin\alpha\, C_\varphi(\alpha, \lambda'). \tag{138}$$

Because $A(\alpha)$ contains only a constant term and second angular harmonics, its symmetric difference is exactly

$$A_+ - A_- = \sin(2h_\varphi) A'(\varphi_c) = \Delta\varphi\ \mathrm{sinc}(2h_\varphi) A'(\varphi_c). \tag{139}$$

For completeness, the corresponding average can be evaluated without forming two nearly equal products:

$$\begin{aligned} \overline{A} &:= \frac{A_+ + A_-}{2} \\ &= \frac{\sin\varphi}{2} + \frac{\cos(2h_\varphi)}{2}\left[\sin\varphi\cos(2\varphi_c) - \cos\varphi\cos(\lambda' - \lambda)\sin(2\varphi_c)\right]. \end{aligned} \tag{140}$$

Define

$$\widehat{\overline{W}} = \frac{\widehat{W}_+ + \widehat{W}_-}{2}, \qquad \widehat{W}_c^{[1]} = \widehat{\mathcal{W}}^{[1]}(c_-, c_+), \tag{141}$$

where the last divided difference is with respect to c. Then

$$\widehat{W}_+ - \widehat{W}_- = (c_+ - c_-)\widehat{W}_c^{[1]}. \tag{142}$$

Applying Eq. (119) and substituting Eqs. (137)–(142) gives the exact identity

$$\boxed{\begin{aligned} A_+\widehat{W}_+ - A_-\widehat{W}_- = \Delta\varphi\Big[&\overline{A}\ \mathrm{sinc}(h_\varphi) C_\varphi(\varphi_c, \lambda')\widehat{W}_c^{[1]} \\ &+ \widehat{\overline{W}}\ \mathrm{sinc}(2h_\varphi) A'(\varphi_c)\Big]. \end{aligned}} \tag{143}$$

Combining Eq. (130) with Eq. (106) therefore yields

$$\boxed{I_\varphi = \Delta r\, \Delta\varphi \int_{\lambda_1}^{\lambda_2} K_\varphi(\lambda')\, \mathrm{d}\lambda',} \tag{144}$$

where

$$\boxed{K_\varphi = \overline{A}\ \mathrm{sinc}(h_\varphi) C_\varphi(\varphi_c, \lambda')\widehat{W}_c^{[1]} + \widehat{\overline{W}}\ \mathrm{sinc}(2h_\varphi) A'(\varphi_c).} \tag{145}$$

The small latitude width is now an explicit factor rather than the result of subtracting the upper and lower boundary contributions.

5.5.4. Cancellation-resistant longitude-boundary difference

Let

$$\beta_c = \lambda_c - \lambda, \qquad \beta_\pm = \beta_c \pm h_\lambda, \qquad B_\pm = \sin\beta_\pm. \tag{146}$$

For each integration point φ', set

$$b(\varphi') = \cos\varphi\cos\varphi', \tag{147}$$

$$c_\pm = \sin\varphi\sin\varphi' + b(\varphi')\cos\beta_\pm, \qquad \widehat{W}_\pm = \widehat{\mathcal{W}}(r, c_\pm). \tag{148}$$

The exact averages and differences are

$$\overline{B} = \frac{B_+ + B_-}{2} = \sin\beta_c\cos h_\lambda, \tag{149}$$

$$B_+ - B_- = \Delta\lambda\ \mathrm{sinc}(h_\lambda)\cos\beta_c, \tag{150}$$

$$c_+ - c_- = -\Delta\lambda\ \mathrm{sinc}(h_\lambda) b(\varphi')\sin\beta_c. \tag{151}$$

Define

$$\widehat{\overline{W}} = \frac{\widehat{W}_+ + \widehat{W}_-}{2}, \qquad \widehat{W}_c^{[1]} = \widehat{\mathcal{W}}^{[1]}(c_-, c_+). \tag{152}$$

Using Eq. (119),

$$\begin{aligned} B_+\widehat{W}_+ - B_-\widehat{W}_- &= \overline{B}(c_+ - c_-)\widehat{W}_c^{[1]} + \widehat{\overline{W}}(B_+ - B_-) \\ &= \Delta\lambda\ \mathrm{sinc}(h_\lambda)\left[\widehat{\overline{W}}\cos\beta_c - b(\varphi')\sin^2\beta_c\cos h_\lambda\,\widehat{W}_c^{[1]}\right]. \end{aligned} \tag{153}$$

Substitution into Eq. (109) gives

$$\boxed{I_\lambda = -\cos\varphi\,\Delta r\,\Delta\lambda\int_{\varphi_1}^{\varphi_2} K_\lambda(\varphi')\,\mathrm{d}\varphi',} \tag{154}$$

with

$$\boxed{K_\lambda = \mathrm{sinc}(h_\lambda)\left[\widehat{\overline{W}}\cos\beta_c - \cos\varphi\cos\varphi'\sin^2\beta_c\cos h_\lambda\,\widehat{W}_c^{[1]}\right].} \tag{155}$$

Thus the small longitude width is also explicit.

5.5.5. Cancellation-resistant explicit term and fully scaled form

Let

$$F(a; r) := a^2 m_a(r), \tag{156}$$

and define its divided difference with respect to a by

$$F_a^{[1]}(r_1, r_2; r) := \frac{F(r_2; r) - F(r_1; r)}{r_2 - r_1}, \tag{157}$$

with the derivative limit used when $r_2 \to r_1$. The exact identity

$$\sin\varphi_2 - \sin\varphi_1 = \Delta\varphi\cos\varphi_c\,\mathrm{sinc}(h_\varphi) \tag{158}$$

converts Eq. (73) into

$$\boxed{C_0 = \Delta r\,\Delta\varphi\,\Delta\lambda\,F_a^{[1]}(r_1, r_2; r)\cos\varphi_c\,\mathrm{sinc}(h_\varphi).} \tag{159}$$

The remaining integration intervals can also be mapped to $[-1, 1]$:

$$\lambda' = \lambda_c + h_\lambda\xi, \qquad \varphi' = \varphi_c + h_\varphi\eta. \tag{160}$$

Then Eqs. (144) and (154) become

$$I_\varphi = \frac{\Delta r\,\Delta\varphi\,\Delta\lambda}{2}\int_{-1}^{1} K_\varphi(\lambda_c + h_\lambda\xi)\,\mathrm{d}\xi, \tag{161}$$

$$I_\lambda = -\frac{\cos\varphi\,\Delta r\,\Delta\varphi\,\Delta\lambda}{2}\int_{-1}^{1} K_\lambda(\varphi_c + h_\varphi\eta)\,\mathrm{d}\eta. \tag{162}$$

Therefore Eq. (110) has the exactly equivalent, fully scaled form

Cancellation-resistant small-cell form. For $P \notin \partial\Omega$, the potential can be written as

$$\boxed{\begin{aligned} V = \frac{G\rho}{2}\,\Delta r\,\Delta\varphi\,\Delta\lambda\Bigg\{ & F_a^{[1]}(r_1, r_2; r)\cos\varphi_c\,\mathrm{sinc}(h_\varphi) \\ & + \frac{1}{2}\int_{-1}^{1} K_\varphi(\lambda_c + h_\lambda\xi)\,\mathrm{d}\xi \\ & - \frac{\cos\varphi}{2}\int_{-1}^{1} K_\lambda(\varphi_c + h_\varphi\eta)\,\mathrm{d}\eta\Bigg\}. \end{aligned}} \tag{163}$$

Here K_φ and K_λ are given by Eqs. (145) and (155). All three small geometric sizes are explicit multiplicative factors.

Equation (163) is algebraically equivalent to Eq. (110); its purpose is numerical stability. The expression is cancellation-resistant in the small-cell limit provided that the radial and angular divided differences are evaluated by the finite local formulas (115)–(116) rather than by direct subtraction of nearly equal endpoint values. The term "cancellation-resistant" is used deliberately: it describes an exact factorization of the geometric small parameters, while the attainable relative accuracy still depends on the evaluation of the remaining smooth functions.

Two elementary component-level rewrites are also useful. For $d_i = d_{r_i}(c)$,

$$d_2 - d_1 = \frac{(r_2 - r_1)(r_1 + r_2 - 2rc)}{d_2 + d_1}, \tag{164}$$

and, for positive A_1 and A_2,

$$\ln\frac{A_2}{A_1} = \mathrm{log1p}\left(\frac{A_2 - A_1}{A_1}\right). \tag{165}$$

These forms should be used when the radial endpoints are close.

The derivative formulas in the following section remain exact. In the small-cell regime, however, $\mathcal{W}$ and its observation-coordinate derivatives should be evaluated through $\mathcal{W} = \Delta r\,\widehat{\mathcal{W}}$, and the paired angular differences should be differentiated in their average–divided-difference form. This prevents analytical differentiation from reintroducing the cancellation that Eq. (163) removes.

6. Exact derivatives through third order of the reduced representation

The potential formula has now reached its intended final form, Eq. (110): one explicit term and two paired one-dimensional integrals. No elliptic-integral reduction is used below. The gradient vector, gravity-gradient tensor, and third-order gravity-gradient tensor (gravitational curvature) are obtained by differentiating this exact reduced representation.

6.0 Why differentiation may be moved inside the integrals

A standard Leibniz differentiation theorem will be used repeatedly. Let

$$I(p) = \int_a^b F(p, \tau)\,\mathrm{d}\tau, \tag{166}$$

where the limits a and b do not depend on p. If F, $\partial F/\partial p$, and the required derivatives through third order are continuous on a neighbourhood of the compact parameter domain, or are bounded

there by an integrable dominating function, then

$$\frac{\partial I}{\partial p} = \int_a^b \frac{\partial F}{\partial p}\,\mathrm{d}\tau, \qquad \frac{\partial^2 I}{\partial p \partial q} = \int_a^b \frac{\partial^2 F}{\partial p \partial q}\,\mathrm{d}\tau. \tag{167}$$

The analogous identity with three partial derivatives follows under the same domination hypothesis. For $P \notin \partial\Omega$, every source point on the four boundary curves has positive distance from the observation point. The reduced kernels are therefore smooth functions of the observation coordinates on each compact integration interval. The conditions of Eq. (167) are satisfied. Consequently, analytical differentiation does not restore any surface or volume integral: the same one-dimensional intervals are retained.

Assume initially that

$$P \notin \partial\Omega, \qquad r \notin \{r_1, r_2\}. \tag{168}$$

The second restriction avoids differentiating the separate piecewise mean terms exactly at a radial source boundary. Where the observation point is not on the actual spherical face, the complete combined expression has a common limiting value as $r \to r_i^{\pm}$; that limit should be taken after terms have been combined.

Removable radial-branch interfaces. Let $r = r_i$ for $i \in \{1, 2\}$ and assume $P \notin \partial\Omega$. Then the angular direction of P lies outside the corresponding closed spherical patch, and there is a neighbourhood U of P with

$$\inf_{\boldsymbol{x} \in U,\, \boldsymbol{y} \in \overline{\Omega}} \|\boldsymbol{x} - \boldsymbol{y}\| > 0. \tag{169}$$

Consequently V is real analytic on U. In particular, the complete reduced representation, after C_0 and all four edge terms have been combined, has identical limits through third order as $r \to r_i^-$ and $r \to r_i^+$.

Justification. The sets consisting of P and $\overline{\Omega}$ are disjoint and compact, so their distance is positive. Shrinking a ball about P by half that distance gives Eq. (169). On the resulting product set the Newton kernel and all of its observation-coordinate derivatives are uniformly bounded. Differentiation under the volume integral is therefore valid to arbitrary order. Since the reduced expression equals that volume integral on both sides of $r = r_i$, uniqueness of limits proves the asserted cancellation of the separate piecewise terms.

This result also explains why one must not differentiate an isolated piecewise mean term at $r = r_i$: its apparent derivative jump is an artefact of the complete-sphere decomposition and cancels only in the assembled formula.

For numerical evaluation it is essential to distinguish this removable radial-branch interface from an actual material boundary. Let the angular projection of P be strictly inside the open patch $(\varphi_1, \varphi_2) \times (\lambda_1, \lambda_2)$. At $r = r_2$ we define all reported quantities by the exterior radial limit

$$X(r_2^+) := \lim_{h \downarrow 0} X(r_2 + h). \tag{170}$$

This convention is evaluated by one-sided Richardson extrapolation. If $r = r_i$ but the angular projection lies outside the closed spherical patch, the point is separated from the source and the equal-radius, cancellation-resistant formula is evaluated directly; no extrapolation is needed. Points on a latitude or longitude edge, points at a corner, and points on the lower material face $r = r_1$ with angular projection inside the patch are not assigned a boundary value here.

They require an independently stated one-sided convention and are therefore rejected by the implementation.

Classical, weak, and distributional scope. The condition $P \notin \partial\Omega$ includes two different classical regimes. If $P \notin \overline{\Omega}$, the potential is harmonic and real analytic near P, and all displayed derivatives can also be obtained by ordinary differentiation of the separated Newton volume kernel. If $P \in \Omega^\circ$, constant density holds in a complete ball about P. Splitting off that ball shows that V is smooth (indeed real analytic) there and satisfies

$$\Delta V = -4\pi G\rho. \tag{171}$$

The reduced boundary formula therefore gives classical derivatives through third order at interior points. However, at such a point one must not obtain the Hessian or third derivatives by naively integrating the pointwise derivatives of $1/l$ across $l = 0$: those raw kernels require the standard principal-value/finite-part interpretation together with their local contact terms. Differentiating the already reduced, boundary-separated representation automatically includes those terms.

On $\partial\Omega$, the potential and its first derivative have their usual traces, but higher classical derivatives generally have side-dependent jumps. The manuscript assigns only the exterior limit on the upper radial face by Eq. (170); all other material faces, edges, and corners are excluded unless a one-sided convention is stated separately. Globally, the unambiguous weak statement is

$$\Delta V = -4\pi G\rho\, \mathbf{1}_\Omega \quad \text{in } \mathcal{D}', \tag{172}$$

and differentiating it once introduces the surface distribution associated with the density jump. Thus the third-order tensor formulas in this section are classical formulas for $P \notin \partial\Omega$ (plus the explicitly selected upper-face exterior limit), not two-sided boundary identities.

Although the observation-radius dependence was suppressed in the potential section, write explicitly

$$\mathcal{H} = \mathcal{H}(r, c), \qquad \mathcal{Q}_a = \mathcal{Q}_a(r, c), \qquad \mathcal{W} = \mathcal{W}(r, c). \tag{173}$$

For differentiation it is algebraically simplest to return to the equivalent four-edge form of Eq. (72). Define

$$B_{\varphi_k} := \epsilon_k \cos\varphi_k\, C_\varphi(\varphi_k, \lambda'), \qquad k = 1, 2, \tag{174}$$

$$B_{\lambda_k} := \epsilon_k C_\lambda(\varphi', \lambda_k), \qquad k = 1, 2. \tag{175}$$

Then

$$V = \frac{G\rho}{2}\left[C_0 + \sum_{e\in\mathcal{E}} \int_{\Gamma_e} B_e \mathcal{W}(r, c)\, \mathrm{d}\tau\right], \qquad \mathcal{E} = \{\varphi_1, \varphi_2, \lambda_1, \lambda_2\}. \tag{176}$$

Equation (176) is exactly equivalent to the paired formula Eq. (110); it is used only because each edge weight can be differentiated separately and then recombined.

6.1 Observation-coordinate derivatives of the angular geometry

For $p, q \in \{r, \varphi, \lambda\}$, use subscripts to denote observation-coordinate partial derivatives. From Eq. (5),

$$c_r = 0, \qquad c_{rr} = c_{r\varphi} = c_{r\lambda} = 0, \tag{177}$$

$$\boxed{c_\varphi = \cos\varphi \sin\varphi' - \sin\varphi \cos\varphi' \cos\beta,} \tag{178}$$

$$\boxed{c_\lambda = \cos\varphi\cos\varphi'\sin\beta,} \tag{179}$$

$$\boxed{c_{\varphi\varphi} = -c,} \tag{180}$$

$$\boxed{c_{\varphi\lambda} = -\sin\varphi\cos\varphi'\sin\beta,} \tag{181}$$

$$\boxed{c_{\lambda\lambda} = -\cos\varphi\cos\varphi'\cos\beta.} \tag{182}$$

The nonzero third angular derivatives are

$$\begin{aligned} c_{\varphi\varphi\varphi} &= -c_\varphi, & c_{\varphi\varphi\lambda} &= -c_\lambda, \\ c_{\varphi\lambda\lambda} &= \sin\varphi\cos\varphi'\cos\beta, & c_{\lambda\lambda\lambda} &= -c_\lambda. \end{aligned} \tag{183}$$

Every third derivative containing an r index vanishes.

The derivatives of the source-latitude edge factor in Eq. (8) are

$$(C_\varphi)_\varphi = \cos\varphi\cos\varphi' + \sin\varphi\sin\varphi'\cos\beta, \tag{184}$$

$$(C_\varphi)_\lambda = -\cos\varphi\sin\varphi'\sin\beta, \tag{185}$$

$$(C_\varphi)_{\varphi\varphi} = -C_\varphi, \tag{186}$$

$$(C_\varphi)_{\varphi\lambda} = \sin\varphi\sin\varphi'\sin\beta, \tag{187}$$

$$(C_\varphi)_{\lambda\lambda} = \cos\varphi\sin\varphi'\cos\beta. \tag{188}$$

Its third derivatives are

$$\begin{aligned} (C_\varphi)_{\varphi\varphi\varphi} &= -(C_\varphi)_\varphi, & (C_\varphi)_{\varphi\varphi\lambda} &= -(C_\varphi)_\lambda, \\ (C_\varphi)_{\varphi\lambda\lambda} &= -\sin\varphi\sin\varphi'\cos\beta, & (C_\varphi)_{\lambda\lambda\lambda} &= -(C_\varphi)_\lambda. \end{aligned} \tag{189}$$

The corresponding derivatives of Eq. (9) are

$$(C_\lambda)_\varphi = \sin\varphi\sin\beta, \tag{190}$$

$$(C_\lambda)_\lambda = \cos\varphi\cos\beta, \tag{191}$$

$$(C_\lambda)_{\varphi\varphi} = -C_\lambda, \tag{192}$$

$$(C_\lambda)_{\varphi\lambda} = -\sin\varphi\cos\beta, \tag{193}$$

$$(C_\lambda)_{\lambda\lambda} = -C_\lambda. \tag{194}$$

Its third derivatives are

$$\begin{aligned} (C_\lambda)_{\varphi\varphi\varphi} &= -(C_\lambda)_\varphi, & (C_\lambda)_{\varphi\varphi\lambda} &= -(C_\lambda)_\lambda, \\ (C_\lambda)_{\varphi\lambda\lambda} &= -(C_\lambda)_\varphi, & (C_\lambda)_{\lambda\lambda\lambda} &= -(C_\lambda)_\lambda. \end{aligned} \tag{195}$$

All radial derivatives of B_e vanish. The angular derivatives of B_e follow directly by multiplying Eqs. (184)–(188) and Eqs. (190)–(194) by the constant prefactors in Eqs. (174) and (175); the same rule applied to Eqs. (189) and (195) supplies $(B_e)_{pqs}$.

6.2 Explicit partial derivatives of the combined kernel

Introduce the physical radial master moment

$$\widehat{M}_n^{(p)}(r,c) := \int_{r_1}^{r_2} \frac{r'^p}{l^n}\,\mathrm{d}r' = r^{p-n+1} M_n^{(p)}(r,c), \tag{196}$$

where $M_n^{(p)}$ is the normalized radial moment used in the preceding derivation. For completeness, define it explicitly by

$$M_n^{(p)}(r,c) := \int_{R_1}^{R_2} \frac{R^p}{(R^2-2Rc+1)^{n/2}}\,\mathrm{d}R, \qquad R_i = \frac{r_i}{r}. \tag{197}$$

The endpoint formulas needed in this work are now given explicitly. Put

$$u = R - c, \qquad s^2 = 1 - c^2, \qquad L = \sqrt{u^2 + s^2}, \tag{198}$$

and, for $|c| < 1$, introduce the primitives

$$J_0^{(3)}(u) = \frac{u}{s^2 L}, \qquad J_1^{(3)}(u) = -\frac{1}{L}, \qquad J_2^{(3)}(u) = \operatorname{asinh}\frac{u}{s} - \frac{u}{L}, \tag{199}$$

$$J_0^{(5)}(u) = \frac{u(3s^2+2u^2)}{3s^4L^3}, \qquad J_1^{(5)}(u) = -\frac{1}{3L^3}, \qquad J_2^{(5)}(u) = \frac{u^3}{3s^2L^3},$$

$$J_3^{(5)}(u) = -\frac{1}{L} + \frac{s^2}{3L^3}. \tag{200}$$

They satisfy

$$\frac{\mathrm{d}}{\mathrm{d}u} J_j^{(n)}(u) = \frac{u^j}{L^n}. \tag{201}$$

Since $R = u + c$, the required normalized moments are the endpoint differences of

$$F_3^{(1)} = J_1^{(3)} + cJ_0^{(3)}, \qquad F_3^{(2)} = J_2^{(3)} + 2cJ_1^{(3)} + c^2 J_0^{(3)}, \tag{202}$$

$$F_5^{(1)} = J_1^{(5)} + cJ_0^{(5)}, \qquad F_5^{(2)} = J_2^{(5)} + 2cJ_1^{(5)} + c^2 J_0^{(5)},$$

$$F_5^{(3)} = J_3^{(5)} + 3cJ_2^{(5)} + 3c^2 J_1^{(5)} + c^3 J_0^{(5)}, \tag{203}$$

namely

$$\boxed{M_n^{(p)} = F_n^{(p)}(u_2) - F_n^{(p)}(u_1), \qquad u_i = R_i - c.} \tag{204}$$

For third order one additionally needs $n = 7$ and $p \le 4$. Rather than listing long primitives, use the executable recurrences

$$J_0^{(n)}(u) = \frac{u}{(n-2)s^2L^{n-2}} + \frac{n-3}{(n-2)s^2} J_0^{(n-2)}(u),$$

$$J_1^{(n)}(u) = -\frac{1}{(n-2)L^{n-2}},$$

$$J_j^{(n)}(u) = J_{j-2}^{(n-2)}(u) - s^2 J_{j-2}^{(n)}(u), \qquad j \ge 2, \tag{205}$$

for odd $n \ge 3$, followed by

$$F_n^{(p)}(u) = \sum_{j=0}^{p} \binom{p}{j} c^{p-j} J_j^{(n)}(u), \qquad M_n^{(p)} = [F_n^{(p)}]_{u_1}^{u_2}. \tag{206}$$

Equations (35), (204), and (205)–(206) close every radial moment used through third order. In floating-point arithmetic the recurrences are applied to endpoint differences, with the small-s branch of Eqs. (208)–(210), so that no large nearly equal primitives are subtracted.

The collinear endpoints are evaluated without dividing by s. If $c = 1$ and the interval does not contain $R = 1$, let

$$\sigma = \operatorname{sgn}(R_1 - 1) = \operatorname{sgn}(R_2 - 1), \qquad v_i = |R_i - 1|. \tag{207}$$

Then

$$\boxed{M_n^{(p)}(r,1) = \left[\sigma \sum_{j=0}^{p} \binom{p}{j} \sigma^j \mathfrak{L}_{j-n}(v)\right]_{v_1}^{v_2},} \tag{208}$$

where

$$\mathfrak{L}_k(v) = \begin{cases} v^{k+1}/(k+1), & k \neq -1, \\ \ln v, & k = -1. \end{cases} \tag{209}$$

For $c = -1$, set $v_i = R_i + 1$; then

$$\boxed{M_n^{(p)}(r,-1) = \left[\sum_{j=0}^{p} \binom{p}{j} (-1)^{p-j} \mathfrak{L}_{j-n}(v)\right]_{v_1}^{v_2}.} \tag{210}$$

Equations (208) and (210) are finite sums of powers and logarithms and therefore give executable endpoint branches. If $c = 1$ and $1 \in [R_1, R_2]$, the moments are singular; geometrically this is precisely the excluded case in which the observation point lies on the radial part of the angular boundary.

Since

$$\mathcal{H}(r,c) = r \int_{r_1}^{r_2} \frac{r'}{l}\, \mathrm{d}r', \tag{211}$$

its required partial derivatives are

$$\boxed{\mathcal{H}_r = r\left[M_1^{(1)} - M_3^{(1)} + cM_3^{(2)}\right],} \tag{212}$$

$$\boxed{\mathcal{H}_c = r^2 M_3^{(2)},} \tag{213}$$

$$\boxed{\mathcal{H}_{rr} = -3M_3^{(1)} + 2cM_3^{(2)} + 3M_5^{(1)} - 6cM_5^{(2)} + 3c^2 M_5^{(3)},} \tag{214}$$

$$\boxed{\mathcal{H}_{rc} = r\left[2M_3^{(2)} - 3M_5^{(2)} + 3cM_5^{(3)}\right],} \tag{215}$$

$$\boxed{\mathcal{H}_{cc} = 3r^2 M_5^{(3)}.} \tag{216}$$

For a compact third-order form, use the physical moments $\widehat{M}_n^{(p)}$ of Eq. (196). They obey

$$\begin{aligned} \partial_r \widehat{M}_n^{(p)} &= -n\left(r\widehat{M}_{n+2}^{(p)} - c\widehat{M}_{n+2}^{(p+1)}\right), \\ \partial_c \widehat{M}_n^{(p)} &= nr\widehat{M}_{n+2}^{(p+1)}. \end{aligned} \tag{217}$$

Consequently,

$$\boxed{\mathcal{H}_{ccc}} = 15r^4\widehat{M}_7^{(4)}, \tag{218}$$

$$\boxed{\mathcal{H}_{rcc}} = 9r^2\widehat{M}_5^{(3)} - 15r^4\widehat{M}_7^{(3)} + 15cr^3\widehat{M}_7^{(4)}, \tag{219}$$

$$\begin{aligned}\boxed{\mathcal{H}_{rrc}} &= 2\widehat{M}_3^{(2)} - 15r^2\widehat{M}_5^{(2)} + 12cr\widehat{M}_5^{(3)} + 15r^4\widehat{M}_7^{(2)} \\ &\quad - 30cr^3\widehat{M}_7^{(3)} + 15c^2r^2\widehat{M}_7^{(4)},\end{aligned} \tag{220}$$

$$\begin{aligned}\boxed{\mathcal{H}_{rrr}} &= -3\widehat{M}_3^{(1)} + 18r^2\widehat{M}_5^{(1)} - 27cr\widehat{M}_5^{(2)} + 9c^2\widehat{M}_5^{(3)} \\ &\quad - 15r^4\widehat{M}_7^{(1)} + 45cr^3\widehat{M}_7^{(2)} - 45c^2r^2\widehat{M}_7^{(3)} + 15c^3r\widehat{M}_7^{(4)}.\end{aligned} \tag{221}$$

These identities include the observation-radius dependence of the normalized endpoints $R_i = r_i/r$; they must therefore be used instead of differentiating $r^2M_1^{(1)}$ while incorrectly holding R_i fixed.

For the gauge coefficient, set

$$L_a := l_a(c), \qquad S_a := (a^2 - r^2)L_a + \frac{1}{3}L_a^3, \qquad \mathcal{P}_a = -\frac{S_a}{2a^2r}. \tag{222}$$

At fixed c,

$$(L_a)_r = \frac{r - ac}{L_a}, \qquad (L_a)_{rr} = \frac{a^2(1 - c^2)}{L_a^3}, \tag{223}$$

$$(L_a)_{rrr} = -\frac{3a^2(1 - c^2)(r - ac)}{L_a^5}. \tag{224}$$

$$(S_a)_r = -2rL_a + (a^2 - r^2 + L_a^2)(L_a)_r, \tag{225}$$

$$(S_a)_{rr} = -2L_a - 4r(L_a)_r + 2L_a(L_a)_r^2 + (a^2 - r^2 + L_a^2)(L_a)_{rr}. \tag{226}$$

$$(S_a)_{rrr} = (a^2 - r^2 + L_a^2)(L_a)_{rrr} - 6r(L_a)_{rr} - 6(L_a)_r + 2(L_a)_r^3 + 6L_a(L_a)_r(L_a)_{rr}. \tag{227}$$

Consequently,

$$\boxed{(\mathcal{P}_a)_r = \frac{S_a}{2a^2r^2} - \frac{(S_a)_r}{2a^2r},} \tag{228}$$

$$\boxed{(\mathcal{P}_a)_{rr} = \frac{1}{2a^2}\left[\frac{2(S_a)_r}{r^2} - \frac{2S_a}{r^3} - \frac{(S_a)_{rr}}{r}\right].} \tag{229}$$

$$\boxed{(\mathcal{P}_a)_{rrr} = \frac{1}{2a^2}\left[-\frac{(S_a)_{rrr}}{r} + \frac{3(S_a)_{rr}}{r^2} - \frac{6(S_a)_r}{r^3} + \frac{6S_a}{r^4}\right].} \tag{230}$$

The derivatives of the complete-sphere mean, away from $r = a$, are

$$(m_a)_r = \begin{cases} -\dfrac{2r}{3a^2}, & r < a, \\ -\dfrac{2a}{3r^2}, & r > a, \end{cases} \qquad (m_a)_{rr} = \begin{cases} -\dfrac{2}{3a^2}, & r < a, \\ \dfrac{4a}{3r^3}, & r > a. \end{cases} \tag{231}$$

and

$$(m_a)_{rrr} = \begin{cases} 0, & r < a, \\ -\dfrac{4a}{r^4}, & r > a. \end{cases} \tag{232}$$

It follows from Eq. (49) that

$$\boxed{(\mathcal{Q}_a)_r = \frac{a^2}{1-c^2}\left[(\mathcal{P}_a)_r(c) - (\mathcal{P}_a)_r(-1) - (m_a)_r(c+1)\right],} \tag{233}$$

$$\boxed{(\mathcal{Q}_a)_{rr} = \frac{a^2}{1-c^2}\left[(\mathcal{P}_a)_{rr}(c) - (\mathcal{P}_a)_{rr}(-1) - (m_a)_{rr}(c+1)\right].} \tag{234}$$

$$\boxed{(\mathcal{Q}_a)_{rrr} = \frac{a^2}{1-c^2}\left[(\mathcal{P}_a)_{rrr}(c) - (\mathcal{P}_a)_{rrr}(-1) - (m_a)_{rrr}(c+1)\right].} \tag{235}$$

The c derivatives are more compactly obtained from the defining differential equation (44). First note that

$$(f_a)_r = -\frac{ar(1-c^2)}{L_a^3}, \qquad (f_a)_c = \frac{r^2(ac-r)}{L_a^3}. \tag{236}$$

The additional derivatives needed below are

$$\begin{aligned}(f_a)_{rr} &= -\frac{a(1-c^2)}{L_a^3} + \frac{3ar(1-c^2)(r-ac)}{L_a^5},\\ (f_a)_{ccc} &= \frac{6a^2r^3}{L_a^5} + \frac{15a^2r^4(ac-r)}{L_a^7}.\end{aligned} \tag{237}$$

Then, for $|c| < 1$,

$$\boxed{(\mathcal{Q}_a)_c = \frac{a^2(f_a - m_a) + 2c\mathcal{Q}_a}{1-c^2},} \tag{238}$$

$$\boxed{(\mathcal{Q}_a)_{cc} = \frac{a^2(f_a)_c + 4c(\mathcal{Q}_a)_c + 2\mathcal{Q}_a}{1-c^2},} \tag{239}$$

$$\boxed{(\mathcal{Q}_a)_{rc} = \frac{a^2[(f_a)_r - (m_a)_r] + 2c(\mathcal{Q}_a)_r}{1-c^2}.} \tag{240}$$

Differentiating these identities once more gives the four symmetric third partials for $|c| < 1$:

$$\boxed{(\mathcal{Q}_a)_{rrc}} = \frac{a^2[(f_a)_{rr} - (m_a)_{rr}] + 2c(\mathcal{Q}_a)_{rr}}{1-c^2}, \tag{241}$$

$$\boxed{(\mathcal{Q}_a)_{rcc}} = \frac{a^2(f_a)_{rc} + 4c(\mathcal{Q}_a)_{rc} + 2(\mathcal{Q}_a)_r}{1-c^2}, \tag{242}$$

$$\boxed{(\mathcal{Q}_a)_{ccc}} = \frac{a^2(f_a)_{cc} + 6c(\mathcal{Q}_a)_{cc} + 6(\mathcal{Q}_a)_c}{1-c^2}. \tag{243}$$

The endpoint limits can be written without division by $1-c^2$. For $a \neq r$ define

$$g_\pm = f_a(\pm 1) - m_a, \qquad f'_\pm = (f_a)_c(\pm 1), \qquad f''_\pm = (f_a)_{cc}(\pm 1), \tag{244}$$

where

$$(f_a)_{cc} = \frac{ar^2}{L_a^3} + \frac{3ar^3(ac-r)}{L_a^5}, \qquad (f_a)_{rc} = \frac{2arc - 3r^2}{L_a^3} + \frac{3r^2(ac-r)^2}{L_a^5}. \tag{245}$$

Expansion of the defining integral at its two endpoints gives

$$(\mathcal{Q}_a)_c(-1) = \frac{a^2}{4}(g_- + f'_-), \qquad (\mathcal{Q}_a)_{cc}(-1) = a^2\left(\frac{g_-}{4} + \frac{f'_-}{4} + \frac{f''_-}{6}\right), \tag{246}$$

$$(\mathcal{Q}_a)_c(1) = \frac{a^2}{4}(g_+ - f'_+), \qquad (\mathcal{Q}_a)_{cc}(1) = -a^2\left(\frac{g_+}{4} - \frac{f'_+}{4} + \frac{f''_+}{6}\right). \tag{247}$$

The radial and mixed endpoint derivatives, again for $a \neq r$, are

$$(\mathcal{Q}_a)_r(-1) = -\frac{a^2}{2}(m_a)_r, \qquad (\mathcal{Q}_a)_{rr}(-1) = -\frac{a^2}{2}(m_a)_{rr},$$

$$(\mathcal{Q}_a)_{rc}(-1) = \frac{a^2}{4}\left[(f_a)_r(-1) - (m_a)_r + (f_a)_{rc}(-1)\right], \tag{248}$$

$$(\mathcal{Q}_a)_r(1) = \frac{a^2}{2}(m_a)_r, \qquad (\mathcal{Q}_a)_{rr}(1) = \frac{a^2}{2}(m_a)_{rr},$$

$$(\mathcal{Q}_a)_{rc}(1) = \frac{a^2}{4}\left[(f_a)_r(1) - (m_a)_r - (f_a)_{rc}(1)\right]. \tag{249}$$

All endpoint derivatives through total order three are covered by one executable Taylor rule. For $a \neq r$ define

$$g^{\pm}_{mk} := \partial_r^m \partial_c^k (f_a - m_a)\Big|_{c=\pm 1}. \tag{250}$$

Then, for $m + n \leq 3$,

$$\boxed{\partial_r^m \partial_c^n \mathcal{Q}_a|_{c=-1}} = a^2 n! \sum_{k=0}^{n} \frac{g^-_{mk}}{(k+1)!\, 2^{n-k+1}}, \tag{251}$$

$$\boxed{\partial_r^m \partial_c^n \mathcal{Q}_a|_{c=1}} = (-1)^{n+1} a^2 n! \sum_{k=0}^{n} \frac{(-1)^k g^+_{mk}}{(k+1)!\, 2^{n-k+1}}. \tag{252}$$

These finite sums follow by expanding the defining integral in $x = c + 1$ or $y = 1 - c$ before division by $1 - c^2$; they include the earlier first- and second-order endpoint formulas and give $Q_{rrr}, Q_{rrc}, Q_{rcc}, Q_{ccc}$ without a small denominator. The quantities $g^{\pm}_{mk}$ are most directly evaluated as a bivariate Taylor jet of the elementary expression for f_a and the piecewise polynomial/rational m_a. When $a = r$ and $c = 1$, Eq. (53) shows that the c derivatives are genuinely singular rather than removable; that configuration is a point of the spherical source face and is excluded.

The partial derivatives of the combined kernel are finally

$$\mathcal{W}_r = (\mathcal{Q}_{r_2})_r - (\mathcal{Q}_{r_1})_r - \mathcal{H}_r, \tag{253}$$

$$\mathcal{W}_c = (\mathcal{Q}_{r_2})_c - (\mathcal{Q}_{r_1})_c - \mathcal{H}_c, \tag{254}$$

$$\mathcal{W}_{rr} = (\mathcal{Q}_{r_2})_{rr} - (\mathcal{Q}_{r_1})_{rr} - \mathcal{H}_{rr}, \tag{255}$$

$$\mathcal{W}_{rc} = (\mathcal{Q}_{r_2})_{rc} - (\mathcal{Q}_{r_1})_{rc} - \mathcal{H}_{rc}, \tag{256}$$

$$\mathcal{W}_{cc} = (\mathcal{Q}_{r_2})_{cc} - (\mathcal{Q}_{r_1})_{cc} - \mathcal{H}_{cc}. \tag{257}$$

The four third partials are assembled by exactly the same signed endpoint difference:

$$\begin{aligned}
\mathcal{W}_{rrr} &= (\mathcal{Q}_{r_2})_{rrr} - (\mathcal{Q}_{r_1})_{rrr} - \mathcal{H}_{rrr},\\
\mathcal{W}_{rrc} &= (\mathcal{Q}_{r_2})_{rrc} - (\mathcal{Q}_{r_1})_{rrc} - \mathcal{H}_{rrc},\\
\mathcal{W}_{rcc} &= (\mathcal{Q}_{r_2})_{rcc} - (\mathcal{Q}_{r_1})_{rcc} - \mathcal{H}_{rcc},\\
\mathcal{W}_{ccc} &= (\mathcal{Q}_{r_2})_{ccc} - (\mathcal{Q}_{r_1})_{ccc} - \mathcal{H}_{ccc}.
\end{aligned} \tag{258}$$

6.3 Spherical-coordinate derivatives through third order

For $p \in \{r, \varphi, \lambda\}$, define the total observation-coordinate derivative of the kernel by

$$D_p \mathcal{W} := \delta_{pr} \mathcal{W}_r + c_p \mathcal{W}_c, \tag{259}$$

where δ_{pr} is the Kronecker delta. For $p, q \in \{r, \varphi, \lambda\}$, define

$$\begin{aligned} D_{pq} \mathcal{W} := {} & \delta_{pr}\delta_{qr} \mathcal{W}_{rr} + (\delta_{pr} c_q + \delta_{qr} c_p) \mathcal{W}_{rc} \\ & + c_p c_q \mathcal{W}_{cc} + c_{pq} \mathcal{W}_c. \end{aligned} \tag{260}$$

For $p, q, s \in \{r, \varphi, \lambda\}$, the third total derivative is

$$\begin{aligned} D_{pqs} \mathcal{W} = {} & \delta_{pr}\delta_{qr}\delta_{sr} \mathcal{W}_{rrr} + \big(\delta_{pr}\delta_{qr} c_s + \delta_{pr}\delta_{sr} c_q + \delta_{qr}\delta_{sr} c_p\big) \mathcal{W}_{rrc} \\ & + \big(\delta_{pr} c_q c_s + \delta_{qr} c_p c_s + \delta_{sr} c_p c_q\big) \mathcal{W}_{rcc} + c_p c_q c_s \mathcal{W}_{ccc} \\ & + \big(\delta_{pr} c_{qs} + \delta_{qr} c_{ps} + \delta_{sr} c_{pq}\big) \mathcal{W}_{rc} \\ & + \big(c_{pq} c_s + c_{ps} c_q + c_{qs} c_p\big) \mathcal{W}_{cc} + c_{pqs} \mathcal{W}_c. \end{aligned} \tag{261}$$

The explicit term depends only on r:

$$(C_0)_r = \Omega_\Gamma \left[r_2^2 (m_{r_2})_r - r_1^2 (m_{r_1})_r \right], \tag{262}$$

$$(C_0)_{rr} = \Omega_\Gamma \left[r_2^2 (m_{r_2})_{rr} - r_1^2 (m_{r_1})_{rr} \right], \tag{263}$$

$$(C_0)_{rrr} = \Omega_\Gamma \left[r_2^2 (m_{r_2})_{rrr} - r_1^2 (m_{r_1})_{rrr} \right], \tag{264}$$

and all of its angular derivatives vanish.

Differentiating Eq. (176) once gives

$$\boxed{V_p = \frac{G\rho}{2} \left[(C_0)_p + \sum_{e \in \mathcal{E}} \int_{\Gamma_e} \left((B_e)_p \mathcal{W} + B_e D_p \mathcal{W} \right) \mathrm{d}\tau \right].} \tag{265}$$

Differentiating a second time gives the symmetric formula

$$\boxed{\begin{aligned} V_{pq} = \frac{G\rho}{2} \Bigg[(C_0)_{pq} + \sum_{e \in \mathcal{E}} \int_{\Gamma_e} \Big(& (B_e)_{pq} \mathcal{W} + (B_e)_p D_q \mathcal{W} \\ & + (B_e)_q D_p \mathcal{W} + B_e D_{pq} \mathcal{W} \Big) \mathrm{d}\tau \Bigg]. \end{aligned}} \tag{266}$$

The symmetric third derivative is

$$\boxed{\begin{aligned} V_{pqs} = \frac{G\rho}{2} \Bigg[(C_0)_{pqs} + \sum_{e \in \mathcal{E}} \int_{\Gamma_e} \Big(& (B_e)_{pqs} \mathcal{W} \\ & + (B_e)_{pq} D_s \mathcal{W} + (B_e)_{ps} D_q \mathcal{W} + (B_e)_{qs} D_p \mathcal{W} \\ & + (B_e)_p D_{qs} \mathcal{W} + (B_e)_q D_{ps} \mathcal{W} + (B_e)_s D_{pq} \mathcal{W} + B_e D_{pqs} \mathcal{W} \Big) \mathrm{d}\tau \Bigg]. \end{aligned}} \tag{267}$$

Equations (265) and (266) show that the potential, all three first derivatives, all six independent second derivatives, and all ten independent third derivatives require the same four one-dimensional

integration intervals. At each quadrature node, the scalar kernel and its nine partial derivatives in Eqs. (253)–(258) can be reused for all components.

6.4 Conversion to the local north–east–radial gradient and Hessian

Let (x, y, z) denote the observation-centred orthonormal frame with x directed north, y east, and z radially outward. The differential displacement in geocentric latitude coordinates is

$$\mathrm{d}\boldsymbol{s} = \boldsymbol{e}_x\, r\,\mathrm{d}\varphi + \boldsymbol{e}_y\, r\cos\varphi\,\mathrm{d}\lambda + \boldsymbol{e}_z\,\mathrm{d}r. \tag{268}$$

Therefore the three orthogonal scale factors are

$$h_x = r, \qquad h_y = r\cos\varphi, \qquad h_z = 1. \tag{269}$$

For any scalar field, the component of its gradient along an orthonormal coordinate direction is the corresponding coordinate derivative divided by the scale factor. Under the sign convention of Eq. (7), the local gradient components are

$$\boxed{V_x = \frac{1}{r}V_\varphi, \qquad V_y = \frac{1}{r\cos\varphi}V_\lambda, \qquad V_z = V_r.} \tag{270}$$

The physical second-order gradient tensor is the covariant Hessian, not merely the matrix of second coordinate derivatives. The reason is that the local unit vectors $\boldsymbol{e}_x$, $\boldsymbol{e}_y$, and $\boldsymbol{e}_z$ vary with position. When the first derivative is differentiated again, derivatives of these basis vectors generate the connection terms proportional to V_r, V_φ, and V_λ. Using the scale factors in Eq. (269), or equivalently the Christoffel symbols of spherical coordinates, gives the following six independent components:

$$\boxed{V_{xx} = \frac{1}{r^2}V_{\varphi\varphi} + \frac{1}{r}V_r,} \tag{271}$$

$$\boxed{V_{yy} = \frac{1}{r^2\cos^2\varphi}V_{\lambda\lambda} + \frac{1}{r}V_r - \frac{\tan\varphi}{r^2}V_\varphi,} \tag{272}$$

$$\boxed{V_{zz} = V_{rr},} \tag{273}$$

$$\boxed{V_{xy} = \frac{1}{r^2\cos\varphi}\left(V_{\varphi\lambda} + \tan\varphi\, V_\lambda\right),} \tag{274}$$

$$\boxed{V_{xz} = \frac{1}{r}V_{r\varphi} - \frac{1}{r^2}V_\varphi,} \tag{275}$$

$$\boxed{V_{yz} = \frac{1}{r\cos\varphi}V_{r\lambda} - \frac{1}{r^2\cos\varphi}V_\lambda.} \tag{276}$$

Thus

$$\boxed{\boldsymbol{T} = \begin{bmatrix} V_{xx} & V_{xy} & V_{xz} \\ V_{xy} & V_{yy} & V_{yz} \\ V_{xz} & V_{yz} & V_{zz} \end{bmatrix}.} \tag{277}$$

The connection terms proportional to V_r, V_φ, and V_λ are essential; omitting them would produce coordinate second derivatives rather than the physical tensor in an orthonormal frame.

For a point outside the source, the tensor satisfies

$$V_{xx} + V_{yy} + V_{zz} = 0. \tag{278}$$

For a point strictly inside the homogeneous prism, it satisfies

$$V_{xx} + V_{yy} + V_{zz} = -4\pi G\rho. \tag{279}$$

These identities provide direct implementation checks. The interior Poisson contribution is already contained in the piecewise complete-sphere terms and the combined edge kernel; no additional correction may be appended to Eqs. (265) or (266).

At the geographic poles, the factors $1/\cos\varphi$ reflect a coordinate singularity rather than a physical singularity. Polar evaluation should use a nonsingular local Cartesian frame or a joint limiting expansion of the numerators and denominators in Eqs. (270)–(276).

6.5 Coordinate-free Cartesian derivatives and the poles

The polar prescription can be made explicit without taking quotients by $\cos\varphi$. Let $\boldsymbol{x} \neq \mathbf{0}$ be the Cartesian observation vector,

$$r = \|\boldsymbol{x}\|, \qquad \boldsymbol{n} = \frac{\boldsymbol{x}}{r}, \tag{280}$$

and let $\boldsymbol{u} = \boldsymbol{e}_{r'}$ be the source radial unit vector at an edge quadrature point. Then $c = \boldsymbol{n} \cdot \boldsymbol{u}$. With Cartesian indices and the summation convention suppressed, define

$$r_i = n_i, \qquad r_{ij} = \frac{\delta_{ij} - n_i n_j}{r}, \tag{281}$$

$$c_i = \frac{u_i - c n_i}{r}, \qquad c_{ij} = \frac{-u_i n_j - u_j n_i - c\delta_{ij} + 3c n_i n_j}{r^2}. \tag{282}$$

Put $P_{ij} = \delta_{ij} - n_i n_j$. The required third geometrical derivatives are

$$r_{ijk} = \frac{-\delta_{ij} n_k - \delta_{ik} n_j - \delta_{jk} n_i + 3 n_i n_j n_k}{r^2}, \tag{283}$$

and

$$\begin{aligned} c_{ijk} = \frac{1}{r^3}\Big[&- u_i P_{jk} - u_j P_{ik} - (u_k - c n_k)\delta_{ij} + 3(u_k - c n_k) n_i n_j \\ &+ 3c(P_{ik} n_j + n_i P_{jk}) - 2 n_k(-u_i n_j - u_j n_i - c\delta_{ij} + 3c n_i n_j)\Big]. \end{aligned} \tag{284}$$

Although Eq. (284) is written with a distinguished index k, its value is fully symmetric; averaging its three cyclic forms is a useful round-off-level symmetry enforcement. For $K(\boldsymbol{x}) = \mathcal{W}(r, c)$, the chain rule gives

$$K_i = \mathcal{W}_r n_i + \mathcal{W}_c c_i, \tag{285}$$

$$\begin{aligned} K_{ij} = {} & \mathcal{W}_{rr} n_i n_j + \mathcal{W}_{rc}(n_i c_j + n_j c_i) + \mathcal{W}_{cc} c_i c_j \\ & + \frac{\mathcal{W}_r}{r}(\delta_{ij} - n_i n_j) + \mathcal{W}_c c_{ij}. \end{aligned} \tag{286}$$

The third derivative is

$$\begin{aligned} K_{ijk} = {} & \mathcal{W}_{rrr} r_i r_j r_k + \mathcal{W}_{rrc}(r_i r_j c_k + r_i r_k c_j + r_j r_k c_i) \\ & + \mathcal{W}_{rcc}(r_i c_j c_k + r_j c_i c_k + r_k c_i c_j) + \mathcal{W}_{ccc} c_i c_j c_k \\ & + \mathcal{W}_{rr}(r_{ij} r_k + r_{ik} r_j + r_{jk} r_i) \\ & + \mathcal{W}_{rc}(r_{ij} c_k + r_{ik} c_j + r_{jk} c_i + c_{ij} r_k + c_{ik} r_j + c_{jk} r_i) \\ & + \mathcal{W}_{cc}(c_{ij} c_k + c_{ik} c_j + c_{jk} c_i) + \mathcal{W}_r r_{ijk} + \mathcal{W}_c c_{ijk}. \end{aligned} \tag{287}$$

Each edge weight can be written invariantly as

$$B_e = \alpha_e \, \boldsymbol{n} \cdot \boldsymbol{t}_e, \tag{288}$$

where

$$(\alpha_e, \boldsymbol{t}_e) = \begin{cases} (\epsilon_k \cos\varphi_k, \boldsymbol{e}_{\varphi'}), & e = \varphi_k, \\ (\epsilon_k, \boldsymbol{e}_{\lambda'}), & e = \lambda_k. \end{cases} \tag{289}$$

If $b_e = \boldsymbol{n} \cdot \boldsymbol{t}_e$, then

$$(B_e)_i = \frac{\alpha_e}{r}(t_{e,i} - b_e n_i), \tag{290}$$

$$(B_e)_{ij} = \frac{\alpha_e}{r^2}\left[-t_{e,i} n_j - t_{e,j} n_i - b_e \delta_{ij} + 3 b_e n_i n_j\right]. \tag{291}$$

All third edge-weight derivatives follow without new trigonometry: define $b_{e,i}$, $b_{e,ij}$, and $b_{e,ijk}$ by Eqs. (282) and (284) after replacing $(\boldsymbol{u}, c)$ by $(\boldsymbol{t}_e, b_e)$, and set

$$(B_e)_{ijk} = \alpha_e b_{e,ijk}. \tag{292}$$

Finally, writing $A(r) = C_0(r)$, the complete Cartesian derivatives are

$$\boxed{V_i = \frac{G\rho}{2}\left[A' n_i + \sum_{e\in\mathcal{E}} \int_{\Gamma_e} \left((B_e)_i \mathcal{W} + B_e K_i\right) \mathrm{d}\tau\right],} \tag{293}$$

$$\boxed{\begin{aligned} V_{ij} = \frac{G\rho}{2}\Bigg[& A'' n_i n_j + \frac{A'}{r}(\delta_{ij} - n_i n_j) \\ & + \sum_{e\in\mathcal{E}} \int_{\Gamma_e} \Big((B_e)_{ij} \mathcal{W} + (B_e)_i K_j + (B_e)_j K_i + B_e K_{ij}\Big) \mathrm{d}\tau\Bigg]. \end{aligned}} \tag{294}$$

Writing $A''' = (C_0)_{rrr}$, the complete third-order Cartesian tensor is

$$\boxed{\begin{aligned} V_{ijk} = \frac{G\rho}{2}\Bigg[& A''' r_i r_j r_k + A''(r_{ij} r_k + r_{ik} r_j + r_{jk} r_i) + A' r_{ijk} \\ & + \sum_{e\in\mathcal{E}} \int_{\Gamma_e} \Big((B_e)_{ijk} \mathcal{W} + (B_e)_{ij} K_k + (B_e)_{ik} K_j + (B_e)_{jk} K_i \\ & + (B_e)_i K_{jk} + (B_e)_j K_{ik} + (B_e)_k K_{ij} + B_e K_{ijk}\Big) \mathrm{d}\tau\Bigg]. \end{aligned}} \tag{295}$$

These formulas contain no polar denominator. At either geographic pole one sets $\boldsymbol{n} = (0, 0, \pm 1)$ and evaluates them directly. Any desired local horizontal components are then obtained by

choosing two orthonormal vectors in the plane perpendicular to $\boldsymbol{n}$. Different choices are related by a planar rotation; the Cartesian tensors themselves are unique.

Let $\boldsymbol{e}_x, \boldsymbol{e}_y, \boldsymbol{e}_z$ be any right-handed local north–east–radial triad (with any fixed horizontal orientation at a pole). The physical third-order tensor is obtained by projection,

$$\boxed{V_{abc} = e_{a,i} e_{b,j} e_{c,k} V_{ijk}, \qquad a, b, c \in \{x, y, z\}.} \tag{296}$$

It has ten independent components,

$$V_{xxx}, V_{xxy}, V_{xxz}, V_{xyy}, V_{xyz}, V_{xzz}, V_{yyy}, V_{yyz}, V_{yzz}, V_{zzz}. \tag{297}$$

Equation (296) automatically contains all connection terms that would appear in a direct third differentiation of spherical-coordinate components, while remaining regular at the poles.

The Cartesian Newton-kernel identity underlying this construction is

$$\partial_i \partial_j \partial_k \frac{1}{l} = \frac{15 d_i d_j d_k}{l^7} - \frac{3(\delta_{ij} d_k + \delta_{ik} d_j + \delta_{jk} d_i)}{l^5}, \qquad \boldsymbol{d} = \boldsymbol{y} - \boldsymbol{x}, \tag{298}$$

which is the unified third-order tensor kernel used in the reference tesseroid formulation. Thus Eq. (295) evaluates the third derivatives analytically and does not obtain them by finite differencing lower-order fields.

6.6 Summary of the derivative structure

At one quadrature node, a program evaluates

$$\mathcal{W}, \quad \mathcal{W}_r, \quad \mathcal{W}_c, \quad \mathcal{W}_{rr}, \quad \mathcal{W}_{rc}, \quad \mathcal{W}_{cc}, \quad \mathcal{W}_{rrr}, \quad \mathcal{W}_{rrc}, \quad \mathcal{W}_{rcc}, \quad \mathcal{W}_{ccc}, \tag{299}$$

together with the elementary angular factors and their derivatives. These quantities may be reused when several components happen to employ the same quadrature node. On the generic line-integral path, the geometric split is common to the requested components: it is determined by a geometry-based candidate on the integration boundary. The adaptive subdivision and stopping test remain component dependent, because different derivative kernels have different magnitudes and cancellation patterns. A single-component request therefore does not evaluate the other 19 components. No sampled componentwise peak set is required by the present formulation. The separated exact-pole branch is an exception: it integrates longitude analytically and forms the complete symmetric tensor at shared radial–polar quadrature nodes. This avoids cancellation between separately evaluated longitude-edge terms; its numerical construction is described in Sec. 7.

6.7 Verification contract for the derivative system

The number of coupled third-order formulas makes manual inspection insufficient. A conforming implementation is therefore required to pass four independent checks.

1. Symbolically differentiate the elementary definitions of c, C_φ, C_λ, L_a, and the radial moment integral, then simplify against Eqs. (178)–(183) and (212)–(221).
2. Compare every Taylor-jet coefficient through total degree three with high-precision automatic differentiation at separated random points and with high-precision finite differences whose step is varied over a convergence window.
3. Verify the Cartesian Hessian and third tensor under every index permutation. Also check the Laplace/Poisson trace and the three differentiated trace identities in Eq. (330).

4. Exercise the endpoint branches $c = \pm 1$, the removable interface $a = r$ away from coincidence, the thin-cell switching thresholds, and points on both sides of every rejected material boundary.

In particular, independent differentiation of the physical radial integral gives

$$\begin{aligned}\mathcal{H}_{rrc} = 2\widehat{M}_3^{(2)} - 15r^2\widehat{M}_5^{(2)} + 12cr\widehat{M}_5^{(3)} + 15r^4\widehat{M}_7^{(2)} \\ - 30cr^3\widehat{M}_7^{(3)} + 15c^2r^2\widehat{M}_7^{(4)},\end{aligned} \tag{300}$$

with each term occurring exactly once. This identity and the observation formula $c_{\varphi\lambda} = -\sin\varphi\cos\varphi'\sin\beta$ are regression-test requirements, not merely typographical conventions.

The executable CPU regression suite supplied with this manuscript records the following results. The decimal references in the second and third rows were evaluated with 100 decimal digits using the exact double-precision inputs.

Check	measured residual	limit
$\mathcal{H}_{rrc}$: Taylor AD versus independent radial moments	6.95×10^{-16}	2×10^{-10}
$\mathcal{Q}_a$, $r = 1$, $a = 1 + 10^{-8}$, $c = \mathrm{nextafter}(1, 0)$	2.83×10^{-16}	5×10^{-15}
$\mathcal{H}$ stable logarithm: two-negative and mixed-sign endpoint tests	rounded to zero	5×10^{-15}
All r, c Taylor coefficients of $\mathcal{Q}_a$ and $\mathcal{H}$ through total degree three at separated test points	3.33×10^{-16}	2×10^{-12}
All 20 exterior potential/gradient/Hessian/third-tensor outputs	finite; tolerance met	required
Hessian trace and three differentiated trace identities	at most 3.97×10^{-21}	2×10^{-8}

The reference implementation and its verification contract use IEEE 754 double-precision CPU arithmetic.

7. Projection-aware adaptive numerical realization

The analytical reduction leaves two paired one-dimensional integrals. The present implementation uses geometry-based split parameters to initialize the adaptive quadrature. It does not use a general peak-search procedure: the location of a sampled maximum is not needed to define the integration partition, and sign-changing derivative kernels need not have their absolute maximum at the distance minimum.

For each paired family, select the boundary with the smaller latitude offset or wrapped longitude offset and form a split candidate t_0. On a latitude boundary, the candidate is the observation longitude, shifted periodically and restricted to the integration interval. On a longitude boundary, it is $\mathrm{atan2}(\sin\varphi, \cos\varphi\cos(\lambda_e - \lambda))$, restricted to the latitude interval. Latitude circles are small circles, except at the equator; longitude boundaries lie on great circles. The clipped t_0 initializes the partition; it is not assumed to be the exact distance minimum on an arbitrary segment or a componentwise kernel maximum. The angular separation is evaluated without the loss of significance of $\arccos c$ near $c = 1$:

$$\gamma = \mathrm{atan2}\left(\|\boldsymbol{n} \times \boldsymbol{u}(t_0)\|, \boldsymbol{n} \cdot \boldsymbol{u}(t_0)\right). \tag{301}$$

With h_t denoting the half-width of the angular integration interval and δ_r the distance of the observation radius from $[r_1, r_2]$, define the dimensionless geometrical width indicator

$$w_g = \frac{\sqrt{\gamma^2 + (\delta_r/r)^2}}{h_t}. \tag{302}$$

The paired interval is split at t_0, and a cosine map clusters nodes on the resulting pieces next to this point. Ordinary pieces adjacent to the split at one end use a half-cosine map with parameter in $[0, \pi/2]$. If a normalized subinterval is classified as adjacent to the split at both ends, the implementation uses the full cosine map over $[0, \pi]$. These maps preserve the integral and reduce rapid variation in the distance factor. The transformed pieces are evaluated by global adaptive Gauss–Kronrod quadrature: GK21 is used when $w_g < 1.5$ and $\delta_r/r < 0.02$, and GK15 otherwise. The default standalone requests are $\epsilon_{\mathrm{abs}} = 10^{-12}$ and $\epsilon_{\mathrm{rel}} = 10^{-10}$. Each transformed piece is limited to 10,000 kernel evaluations and 500 intervals. Reaching either limit without meeting the stopping test is recorded as nonconvergence, rather than as successful completion.

If I is one component and A is the corresponding numerical approximation to $\int |K|\,\mathrm{d}t$, termination requires

$$E \le \max\left\{\epsilon_{\mathrm{abs}}, \epsilon_{\mathrm{rel}}|I|, 50\epsilon_{\mathrm{mach}}A\right\}. \tag{303}$$

Thus a nearly vanishing, cancellation-dominated component cannot be declared accurate merely because its computed value is small. Such components are assessed primarily by absolute error. In IEEE 754 double-precision arithmetic the requested relative tolerance is not allowed to fall below $500\epsilon_{\mathrm{mach}}$, because a smaller value causes the embedded Gauss–Kronrod difference to chase round-off without supplying reproducible digits. This lower bound does not replace the componentwise absolute-error test in Eq. (303).

The projection split is a geometric safeguard rather than a peak-localization theorem. Validation should include near-edge, near-corner, odd-kernel, and displaced-observation cases. Acceptance depends on the componentwise error estimate, not on a sampled peak amplitude. The requested tolerances and Eq. (303) control the embedded quadrature estimate; they do not constitute a bound on the total floating-point or source-model error.

Upper-face limit and separated exact-pole branch. Ordinary off-boundary line integration uses the Gauss–Kronrod procedure above. One-sided Richardson extrapolation is used separately for the prescribed upper-face exterior limit in Eq. (170). At an exact geographic pole separated from the source either radially or angularly, the current implementation instead evaluates the Newtonian Cartesian kernels after analytical longitude integration. Their longitude dependence through third order consists of elementary trigonometric moments of degree at most three. The remaining radial–polar integral is evaluated by a double-exponential product rule, with shared nodes for all 20 quantities.

Let θ be the angular distance from the observation pole and a the source radius. The distance-resolving change of variables is

$$\eta = 2\sqrt{ra}\sin(\theta/2), \qquad l^2 = (r-a)^2 + \eta^2, \qquad a^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}a = \frac{a\eta}{r}\,\mathrm{d}\eta\,\mathrm{d}a. \tag{304}$$

For $a \neq r$, a further map $\eta = |r-a|\sinh s$ resolves a small radial gap without subtracting large longitude-edge contributions. If a radial node has $a = r$ while the source is angularly separated from the pole, direct integration in η is regular. When $r \in (r_1, r_2)$ and the source is angularly separated, the radial interval is split at $a = r$. Successive product-rule levels provide the convergence estimate. This is a specialized numerical path for separated polar observations;

the gauge-line identity itself remains valid there, and the branch does not introduce a general interior or two-sided boundary formula.

7.1 Hybrid organization used by TLI-MMSI

Mass and Magnetization Shell Integration (MMSI) stores density or magnetization and the upper and lower shell radii on a regular, cell-centred latitude–longitude grid. TLI–MMSI uses these data in two source representations. The far region integrates the continuous spherical bilinear interpolant, which is linear in longitude and in $\sin\varphi'$. The near region first samples that interpolant on an observation-centred grid, then uses a homogeneous central cell and interpolated states on its surrounding grid lines. The two regions share the underlying data, but they are not exact integrals of the same continuous source model. The homogeneous-tesseroid identity of Sec. 5 supplies the kernel; the shell realization also includes regridding and source approximation.

Let the source-grid spacings be $\Delta\lambda$ and $\Delta\varphi$. The observation-centred cell is denoted by H_0. Define the near-field half-widths

$$h_\lambda = \max(1^\circ, \Delta\lambda/2), \qquad h_\varphi = \max(1^\circ, \Delta\varphi/2), \tag{305}$$

and the rectangle

$$H_1 = [\lambda - h_\lambda, \lambda + h_\lambda] \times [\max(-90^\circ, \varphi - h_\varphi), \min(90^\circ, \varphi + h_\varphi)]. \tag{306}$$

Longitude is unwrapped about the observation longitude before this rectangle The rectangles H_0 and H_1 are understood after intersection with the full source domain D; the full angular source domain is therefore the disjoint union

$$D = (D \cap H_0) \,\dot\cup\, ((D \cap H_1) \setminus H_0) \,\dot\cup\, (D \setminus H_1). \tag{307}$$

For any selected component T_α, the numerical hybrid approximation is assembled as

$$T_\alpha = T_\alpha^{H_0,\mathrm{TLI}} + T_\alpha^{H_1\setminus H_0,\text{agg-TLI}} + T_\alpha^{D\setminus H_1,\mathrm{2D}}. \tag{308}$$

The H_0 term uses one locally sampled material state and one pair of radial bounds in the homogeneous-tesseroid formula, evaluated by the adaptive procedure above. In the implementation used here, its effective absolute and relative requests are both bounded below by 10^{-10}. This hybrid floor is distinct from the standalone relative floor $500\epsilon_{\mathrm{mach}}$ in Sec. 7; an input request of 10^{-16} is therefore not the effective tolerance of every hybrid stage.

For $R = H_1 \setminus H_0$, spherical-face terms are accumulated explicitly and edges on a common grid line are combined before integration. To state this assembly without counting the orientation twice, write $B_e = s_e b_e$, where $s_e = \pm 1$ is the outward orientation and $b_e = \cos\varphi_e C_\varphi$ on a latitude edge or $b_e = C_\lambda$ on a longitude edge. For a fixed collection of homogeneous cells, the assembly is

$$\begin{aligned} \mathcal{S}_{R,\alpha} &= \sum_c \rho_c \, \partial_\alpha C_{0,c}^R, \\ \mathcal{A}_{j,\alpha}(t) &= \sum_{e\subset L_j} s_e \rho_e(t) \, \partial_\alpha [b_e(t) \mathcal{W}_e(t)] \,, \\ T_\alpha^{R,\text{agg-TLI}} &= \frac{G}{2} \left[\mathcal{S}_{R,\alpha} + \sum_j \int_{L_j \cap R} \mathcal{A}_{j,\alpha}(t) \, \mathrm{d}t \right]. \end{aligned} \tag{309}$$

Here $C^R_{0,c}$ is the explicit spherical-face term of cell c, weighted by the solid angle of its part in R; for the potential, ∂_α is the identity. Observation derivatives act on the gravitational kernel with source geometry and material states held fixed, followed by projection onto the reported local basis. Density is displayed explicitly, so $\mathcal{W}_e$ retains the length-squared definition in Eq. (71). The same assembly structure is used for the sampled hybrid model, with row or column interpolation supplying the edge states. That interpolation is a source approximation, not a new exact reduction for a continuously varying density or interface. The resulting fields are componentwise approximations to the derivatives of the prescribed shell model. Since the near reconstruction depends on the observation point, derivative consistency of the assembled hybrid potential requires a separate check; the kernel differentiation above does not differentiate the reconstruction or partition.

Shared interior edges with identical states cancel algebraically; material or radial-boundary changes remain as oriented jumps. The explicit term $\mathcal{S}_{R,\alpha}$ does not disappear in this cancellation. Selected lines are split at source interpolation knots. Fine-grid panels use GL2, coarser panels GL4, and panels next to the geometric split point use a cosine-clustered GL10 rule. Compensated summation is applied to the edge-state differences and to the final component totals.

Write the bounds of H_1 as $(\lambda_W, \lambda_E, \varphi_S, \varphi_N)$ in the unwrapped observation-centred longitude system. The far field consists of four non-overlapping rectangles,

$$\begin{aligned} R_S &= [\varphi_D^S, \varphi_S] \times [\lambda - 180^\circ, \lambda + 180^\circ], \\ R_N &= [\varphi_N, \varphi_D^N] \times [\lambda - 180^\circ, \lambda + 180^\circ], \\ R_W &= [\varphi_S, \varphi_N] \times [\lambda - 180^\circ, \lambda_W], \\ R_E &= [\varphi_S, \varphi_N] \times [\lambda_E, \lambda + 180^\circ], \end{aligned} \tag{310}$$

where φ_D^S and φ_D^N are the southern and northern limits of the source grid. Empty rectangles are omitted. Analytical radial integration provides a two-dimensional surface kernel $K_\alpha^{(2)}$, so

$$T_\alpha^{D\setminus H_1,\mathrm{2D}} = \sum_{q\in\{S,N,W,E\}} \iint_{R_q} K_\alpha^{(2)}(r,\varphi,\lambda;\varphi',\lambda')\,\mathrm{d}\varphi'\mathrm{d}\lambda'. \tag{311}$$

The source fields inside $K_\alpha^{(2)}$ are evaluated directly from the spherical bilinear interpolant of the original source grid. Excluding H_1 removes the sharply localized distance factor from this stage, although interpolation knots still affect source regularity. The implementation uses an embedded 14×14 rule without projection splitting or cosine clustering and permits at most 500 rule evaluations in each far rectangle. Component masks are propagated to this stage, so a V_z-only or magnetic-vector calculation does not evaluate unused surface kernels. The quadrature routines retain error estimates and evaluation-limit diagnostics; the existence of a finite returned value alone is not a certificate that its requested tolerance was met.

When both $\Delta\lambda \geq 2^\circ$ and $\Delta\varphi \geq 2^\circ$, the central cell fills H_1 (with the same latitude clipping), and Eq. (308) reduces to adaptive TLI on H_0 plus two-dimensional integration outside it. If only one spacing meets this condition, the aggregated transition region can remain in the other direction. For finer grids, $H_1 \setminus H_0$ supplies the surrounding near region.

The partition omits no part of the source domain. Its error contributions can nevertheless be separated schematically as

$$E_{\mathrm{total}} \lesssim E_{\mathrm{model}} + E_{H_0}^{\mathrm{TLI}} + E_{H_1}^{\mathrm{agg}} + E_{\mathrm{far}}^{\mathrm{2D}} + E_{\mathrm{switch}} + E_{\mathrm{fp}}. \tag{312}$$

E_{model} includes the source-grid representation, near-grid resampling, and homogeneous central-cell approximation; $E_{H_0}^{\mathrm{TLI}}$, $E_{H_1}^{\mathrm{agg}}$, and $E_{\mathrm{far}}^{\mathrm{2D}}$ are the three integration errors; E_{switch} describes the mismatch between near and far source representations and their treatment at ∂H_1; and E_{fp} is the floating-point error. Equation (312) is an error bookkeeping relation, not a proved computable bound. Increasing quadrature order cannot remove source-model or representation mismatch. This description specifies the current supplied implementation.

7.2 Interpolation and continuity

The underlying spherical bilinear fields are continuous at shared grid nodes, but their first derivatives need not be continuous across source-grid boundaries. The current near-region implementation also smooths the local state and the placement of the central-cell hole in narrow bands around interpolation knots. It blends the neighbouring linear branches with $w(u) = 6u^5 - 15u^4 + 10u^3$, $0 \le u \le 1$, over a coordinate half-width $\min(0.05^\circ, 0.01\Delta)$, where Δ is the corresponding grid spacing. Longitude and latitude hole configurations are combined by their product weights. This local smoothing is applied to the near-cell construction; it does not replace the continuous bilinear far-field source model.

High-order fields can be sensitive to these interpolation and regridding choices even when the potential appears smooth. The exact identity for a homogeneous tesseroid does not by itself prove equality of the two shell representations, smoothness of the assembled approximation across their switch, or convergence of that approximation as the source grid is refined. Those properties require separate numerical checks. They should be distinguished from the dimensional-reduction identity and from convergence of an individual quadrature.

8. Observation point at the centre (theoretical special case)

The following formulas give the continuous theoretical limit at the centre. The current production implementation requires $r > 0$ and therefore does not expose this branch as a general runtime path.

The normalization in Eq. (27) assumes $r > 0$. If $r = 0$, then $l = r'$, and Eq. (7) gives directly

$$\boxed{V(0) = \frac{G\rho}{2}(r_2^2 - r_1^2)(\lambda_2 - \lambda_1)(\sin\varphi_2 - \sin\varphi_1).} \tag{313}$$

This is also the continuous $r \to 0$ limit of Eq. (72); all four one-dimensional terms vanish in that limit.

Although spherical observation coordinates are undefined at the centre, the Cartesian derivatives are regular because $r_1 > 0$. Let

$$\boldsymbol{u}(\varphi', \lambda') = \begin{bmatrix} \cos\varphi'\cos\lambda' \\ \cos\varphi'\sin\lambda' \\ \sin\varphi' \end{bmatrix}, \qquad \mathrm{d}\omega = \cos\varphi'\,\mathrm{d}\varphi'\mathrm{d}\lambda', \tag{314}$$

and define the angular moments

$$\boldsymbol{A}_\Gamma = \int_\Gamma \boldsymbol{u}\,\mathrm{d}\omega, \qquad \boldsymbol{B}_\Gamma = \int_\Gamma \boldsymbol{u}\boldsymbol{u}^{\mathsf{T}}\,\mathrm{d}\omega, \qquad (C_\Gamma)_{ijk} = \int_\Gamma u_i u_j u_k\,\mathrm{d}\omega. \tag{315}$$

Differentiating the Newton kernel in Cartesian coordinates gives

$$\boxed{\nabla V(\mathbf{0}) = G\rho(r_2 - r_1)\boldsymbol{A}_\Gamma,} \tag{316}$$

$$\boxed{\nabla\nabla V(\mathbf{0}) = G\rho \ln\frac{r_2}{r_1}\,(3\boldsymbol{B}_\Gamma - \Omega_\Gamma \boldsymbol{I})\,.} \tag{317}$$

The third derivative at the centre is likewise finite and equals

$$\boxed{V_{ijk}(\mathbf{0}) = 3G\rho\left(\frac{1}{r_1}-\frac{1}{r_2}\right)\left[5(C_\Gamma)_{ijk} - (\delta_{ij}A_{\Gamma,k} + \delta_{ik}A_{\Gamma,j} + \delta_{jk}A_{\Gamma,i})\right].} \tag{318}$$

These angular moments are elementary. For a compact closed form, set

$$\begin{aligned} L_c &= \sin\lambda_2 - \sin\lambda_1, & L_s &= \cos\lambda_1 - \cos\lambda_2,\\ L_{cc} &= \frac{\Delta\lambda}{2} + \frac{\sin 2\lambda_2 - \sin 2\lambda_1}{4}, & L_{ss} &= \frac{\Delta\lambda}{2} - \frac{\sin 2\lambda_2 - \sin 2\lambda_1}{4},\\ L_{cs} &= \frac{\cos 2\lambda_1 - \cos 2\lambda_2}{4}, \end{aligned} \tag{319}$$

and

$$\begin{aligned} P_2 &= \left[\frac{\varphi'}{2} + \frac{\sin 2\varphi'}{4}\right]_{\varphi_1}^{\varphi_2}, & P_{sc} &= \frac{\sin^2\varphi_2 - \sin^2\varphi_1}{2},\\ P_3 &= \left[\sin\varphi' - \frac{\sin^3\varphi'}{3}\right]_{\varphi_1}^{\varphi_2}, & P_{2s} &= \frac{\cos^3\varphi_1 - \cos^3\varphi_2}{3},\\ P_{s2} &= \frac{\sin^3\varphi_2 - \sin^3\varphi_1}{3}. \end{aligned} \tag{320}$$

Then

$$\boldsymbol{A}_\Gamma = \begin{bmatrix} P_2 L_c \\ P_2 L_s \\ P_{sc}\Delta\lambda \end{bmatrix}, \qquad \boldsymbol{B}_\Gamma = \begin{bmatrix} P_3 L_{cc} & P_3 L_{cs} & P_{2s} L_c \\ P_3 L_{cs} & P_3 L_{ss} & P_{2s} L_s \\ P_{2s} L_c & P_{2s} L_s & P_{s2}\Delta\lambda \end{bmatrix}. \tag{321}$$

Since $\operatorname{tr}\boldsymbol{B}_\Gamma = \Omega_\Gamma$, the centre Hessian is trace free, as expected because the central ball $r < r_1$ contains no source. The ten entries of $\boldsymbol{C}_\Gamma$ are also elementary products of longitude integrals of $\cos^p\lambda'\sin^q\lambda'$ and latitude integrals of $\cos^p\varphi'\sin^q\varphi'$; evaluating these endpoint antiderivatives is the executable centre branch for the third-order tensor.

9. Complete-spherical-shell check

For

$$\varphi_1 = -\frac{\pi}{2}, \qquad \varphi_2 = \frac{\pi}{2}, \qquad \lambda_2 - \lambda_1 = 2\pi, \tag{322}$$

the four line-integral terms cancel and $\Omega_\Gamma = 4\pi$. Equation (72) reduces to

$$V = 2\pi G\rho\left(r_2^2 m_{r_2} - r_1^2 m_{r_1}\right). \tag{323}$$

If $r > r_2$, Eq. (40) gives

$$V = \frac{4\pi G\rho}{3r}(r_2^3 - r_1^3) = \frac{GM}{r}, \tag{324}$$

where $M = 4\pi\rho(r_2^3 - r_1^3)/3$. If $r < r_1$,

$$V = 2\pi G\rho(r_2^2 - r_1^2), \tag{325}$$

which is the classical interior potential of a homogeneous spherical shell.

The remaining radial regime provides a check of the two different branches of m_{r_1} and m_{r_2}. If $r_1 < r < r_2$, then

$$\boxed{V(r) = 2\pi G\rho \left(r_2^2 - \frac{r^2}{3} - \frac{2r_1^3}{3r} \right).} \tag{326}$$

Its first two radial derivatives are

$$V_r = 2\pi G\rho \left(-\frac{2r}{3} + \frac{2r_1^3}{3r^2} \right), \qquad V_{rr} = 2\pi G\rho \left(-\frac{2}{3} - \frac{4r_1^3}{3r^3} \right). \tag{327}$$

The nonzero independent third-order components in a local frame are $V_{zzz} = V_{rrr}$ and $V_{xxz} = V_{yyz} = V_{rr}/r - V_r/r^2$. Hence, in the three radial regimes,

$$\boxed{(V_{xxz}, V_{yyz}, V_{zzz}) = \begin{cases} (0,0,0), & r < r_1, \\ \left(-\dfrac{4\pi G\rho r_1^3}{r^4}, -\dfrac{4\pi G\rho r_1^3}{r^4}, \dfrac{8\pi G\rho r_1^3}{r^4} \right), & r_1 < r < r_2, \\ \left(\dfrac{3GM}{r^4}, \dfrac{3GM}{r^4}, -\dfrac{6GM}{r^4} \right), & r > r_2. \end{cases}} \tag{328}$$

Every other independent third-order component vanishes by spherical symmetry. Spherical symmetry gives the two tangential Hessian eigenvalues V_r/r; hence

$$V_{rr} + \frac{2}{r} V_r = -4\pi G\rho, \tag{329}$$

which verifies the interior Poisson equation and the branch coupling in the reduced representation. Differentiating the trace gives the three third-order consistency identities away from a density discontinuity,

$$V_{xxx} + V_{xyy} + V_{xzz} = 0, \qquad V_{xxy} + V_{yyy} + V_{yzz} = 0, \qquad V_{xxz} + V_{yyz} + V_{zzz} = 0. \tag{330}$$

They provide stringent componentwise tests of the implementation; the last is also verified directly by Eq. (328).